Scaling of matrix-product-state plus cosine-filter ground-state preparation: gap, overlap and target in the J_1–J_2 model, with a comparison to QETU

Authors to be added
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A common route to a ground state on a quantum computer compiles a classical trial state and removes its residual excited weight with a quantum filter. We study how the cost of this workflow scales, with a matrix-product state (MPS) as the trial and the one-ancilla cosine filter of Stetcu, Baroni and Carlson (SBC) as the refinement, on the spin-$\frac{1}{2}$ J_1–J_2 model. A modular pipeline turns DMRG inputs into a Trotterized circuit with an input-conditional bound on the infidelity and a gate count; the bound held in all 96 simulated circuits. Each controlled Heisenberg bond compiles exactly into 6 CNOTs, half the per-term count. We report two costs throughout: *certified*, the circuit the pipeline plans from the bound, which can be run as is, and *measured*, the cheapest circuit that reaches the target in exact simulation, whose filter length and Trotter step are tuned with knowledge of the ground state and are therefore a lower reference, not a recipe. Holding the spectral gap fixed across size by tuning J_2 and the trial overlap fixed by a sublattice rotation (222 cases, $N = 8$–14, $\epsilon = 10^{-2}$–10^{-4}), the cost factorizes into filter length, Trotter step and gates per step. The gap sets the circuit: the evolution time follows $T \propto \Delta^{-1} \ln[(1-\gamma)/(\gamma\epsilon)]$ independently of N. The overlap sets the repetitions: lowering it thirtyfold raises the measured CNOTs per shot only 1.2–3.0×, but the cost per success by a further $1/\gamma$. Certification sizes the filter to within 2.0× but asks for Trotter steps 60–75 times finer than needed. This overhead decides the comparisons. Measured, the SBC filters beat a Chebyshev (QETU) filter at $\epsilon \geq 10^{-3}$, because their long pulses tolerate Trotter steps of order $1/J_1$; certified, QETU is cheaper than the closed-form SBC schedule in every case and is the cheapest filter in 159 of 222. Measured end to end, filtering a cheap MPS beats compiling a better one on chains only as a last-mile polisher (18 of 63 targets), but on 4×3 and 4×4 cylinders at 27 of 28 targets, by up to 8.8×, and still at 24 of 28 with MPS circuits costed at the isometry lower bound. Certified, the filtered route wins essentially only where no MPS with $\chi \leq 64$ reaches the target (QETU adds one finite win, by 1.2×); closing that gap needs tighter, state-dependent Trotter error bounds rather than better filters.

I. INTRODUCTION

A common route to a ground state on a quantum computer starts classically: an approximate method produces a trial state that can be compiled into a short circuit, and a quantum subroutine removes its residual excited-state weight. For one-dimensional systems the natural classical engine is DMRG [3, 4], which yields the ground-state energy E_0, the gap and a matrix-product state (MPS) whose low-bond-dimension truncations compile exactly into staircase circuits [5]. The quantum step can be phase estimation [6], a polynomial filter implemented by quantum singular-value transformation [7–9], or a product of cosine filters implemented with a single ancilla [1, 13]. The last option, the projection algorithm of Stetcu, Baroni and Carlson (SBC) [1], needs little beyond time evolution: each pulse is one controlled time evolution between ancilla rotations and a measurement, the filter is designed classically, and the success probability is known in advance.

This paper asks three questions about the SBC filter on a concrete, physically relevant model, the open J_1–J_2 Heisenberg chain:

1. *What does it cost?* Two-qubit gates, arbitrary rotations, evolution time and repetitions, both as certified by an a-priori bound and as measured by exact simulation of the Trotterized circuit.

2. *What sets the cost?* The spectral gap Δ and the trial overlap γ normally change together with system size N and with the coupling. We decouple them: the gap is held fixed across N by tuning J_2, and the overlap is held fixed by a one-parameter rotation of the trial circuit, so each can be varied while the other stays put.

3. *When does it pay off?* The alternative to filtering a cheap trial state is compiling a better one. We compare the two routes at equal target infidelity ϵ.

The answers rest on a modular pipeline (Sec. II) that takes energy estimates and an ansatz for any Pauli-sum Hamiltonian, designs a cosine filter with a certified suppression ratio, chooses Trotter step counts from commutator bounds, and returns an a-priori bound on the infidelity together with a gate count. Section III introduces the model and validates the bound. Section IV is the core of the paper: the controlled gap/overlap study, which yields the cost laws and explains them through a factorization into filter length, Trotter step and gates per step. Section V follows the natural chains to $N = 48$, tests robustness to wrong inputs and measures where the filter beats a compiled MPS. Section VI repeats that comparison on $4 \times L_x$ cylinders, where it comes out the other way, and Sec. VII compares with a Chebyshev polynomial filter (QETU) [9] under identical accounting. Section VIII sets every measured comparison beside its certified, implementable counterpart.

Relation to prior work. The filter is SBC's [1]; it shares

its circuit with the rodeo algorithm [13, 20], which draws the times at random. Our closed-form "dyadic" schedule, a halving ladder of times, is the one SBC and Rule *et al.* use for quantum-number projection [1, 21], and for energy it is a finite form of the rodeo super-iteration of Cohen and Oh [22], recently generalized to other geometric ratios [23]; what we add is its certified bound on the scaled spectral window and its measured cost. The optimized schedule is SBC's energy recipe with a certified constraint. The Trotter bounds are those of Childs *et al.* [2], and the certificate follows the standard argument for polynomial filters [7, 9]. QETU has been costed with Trotterized evolution for lattice gauge theories [24] and run on small Hubbard models [25, 26]. Resource studies that compile an MPS or other classical trial and then refine it exist for quantum chemistry [11, 12, 27], and Watanabe *et al.* [28] stop an MPS encoder early and finish with probabilistic imaginary-time evolution on Heisenberg chains. For the two-dimensional J_1–J_2 model, the quantum–classical crossover has been estimated for phase estimation against DMRG with state preparation taken as negligible [29]. What is new here is (i) a controlled separation of gap and overlap at fixed size, (ii) a measured, not only bounded, account of every resource and its factorization, which shows where the certificate's overhead sits, (iii) a like-for-like measured comparison of cosine and Chebyshev filters, and (iv) a measured map of where filtering a cheap MPS beats compiling a better one, in one and two dimensions. The code is the `sbpipe` package, available with all data at `https://github.com/aaronseymour7/FilterPipeline`; every number below is produced by the scripts named in Appendix B.

II. THE SBC FILTER AND THE RESOURCE PIPELINE

A. Inputs and the spectral window

The Hamiltonian is $H = \sum_a c_a P_a$ with real c_a and Pauli strings P_a, in a fixed order that is also the Trotter order, and with an optional grouping of terms (here: one group per bond). The pipeline assumes

$$\begin{aligned} E_0 &\in [E_0^{\mathrm{est}} - \delta E_0,\ E_0^{\mathrm{est}} + \delta E_0], \\ E_1 &\geq E_1^{\mathrm{est}} - \delta E_1, \qquad E_{\mathrm{top}} \leq E_{\max}, \end{aligned} \tag{1}$$

where E_1 is the lowest level above the ground space that the trial state can overlap with. When the trial state's quantum numbers are unknown, as for the DMRG trial states below, which are not symmetry-adapted, this is the global first excited level; restricting to the trial state's symmetry sector could only raise it. The certificate is a statement about the actual Hamiltonian only if (1) holds for it: with an overestimated E_1 a real excitation can sit in the filter's transition region, and the bound then says nothing (Sec. V B). If no $E_{\max}$ is given, the Weyl bound $E_{\max} = \sum_g \lambda_{\max}(h_g)$ over the term groups h_g is used; it is rigorous and, for the antiferromagnetic J_1–J_2 chain, attained by the ferromagnetic state. With $s = E_0^{\mathrm{est}} - \delta E_0$ and $W = E_{\max} - s$, the scaled Hamiltonian $H_s = (H - s)/W$ has its ground energy in the *pass band* $[0, w_0]$, $w_0 = 2\delta E_0/W$, and every other relevant level in the *stop band* $[\Delta, 1]$, $\Delta = (E_1^{\mathrm{est}} - \delta E_1 - s)/W$.

The ansatz enters through γ, a lower estimate of its weight on the ground space, and through the gate cost of the circuit that prepares it. A statevector is needed only for validation.

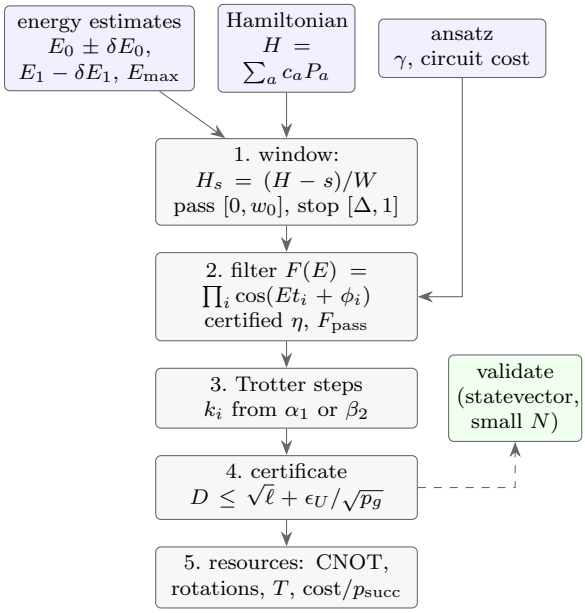


FIG. 1. The pipeline. Blue: inputs, which any classical method can supply. Grey: the stages of `refine()`. Green: the optional statevector check used in Sec. III B.

B. The filter and three schedules

One SBC pulse [1] applies $\mathsf{H}_a\, e^{-itH_s \otimes Z_a}\, R_z^{(a)}(2\phi)\, \mathsf{H}_a$ to the system and an ancilla in $|0\rangle$; the ancilla-$|0\rangle$ block of this unitary is $\cos(H_s t + \phi)$. (SBC write the same block with $e^{-i(tH+\delta)\otimes Y_a}$.) Keeping outcome 0 after m pulses applies

$$F(H_s) = \prod_{i=1}^{m} \cos(H_s t_i + \phi_i), \qquad T = \sum_i |t_i|. \tag{2}$$

Two numbers characterize a filter: the pass-band floor $F_{\mathrm{pass}} \leq \min_{[0,w_0]} |F|$ and the suppression ratio $\eta = \sup_{[\Delta,1]} |F|/F_{\mathrm{pass}}$. Both are certified, not sampled: since $|F'| \leq L := \sum_i |t_i|$ and $|F''| \leq L^2$, an interval branch-and-bound gives a rigorous upper bound on $\sup |F|$ over the stop band, and $F_{\mathrm{pass}} = |F(0)| - Lw_0$ (the certifier is `sbpipe.floor.certify`; it is used for the optimized schedule, the other two have closed-form certificates). Given a target leakage ϵ_ℓ and γ, the design target is

$\eta^2 \le \gamma\epsilon_\ell/[(1-\epsilon_\ell)(1-\gamma)]$ (Lemma 1). We provide three schedules:

Uniform: m equal pulses $t = \pi/(1+\Delta)$, $\phi = 0$, so $F = \cos^m(Et)$. For $E \in [\Delta, 1]$ we have $Et \in [\theta, \pi-\theta]$, $\theta = \pi\Delta/(1+\Delta)$, hence $\sup_{[\Delta,1]} |\cos Et| = \cos\theta$. Then $m \approx 2\ln(1/\eta)/\theta^2$ and $T \approx 2\ln(1/\eta)/(\pi\Delta^2)$: quadratic in the inverse gap. It is the baseline that any design should beat.

Dyadic: r repetitions of the halving ladder $t = \tau, 2\tau, \ldots, 2^{K-1}\tau$, $\tau = \pi/(1+\Delta)$, $\phi = 0$. This is the schedule SBC use for quantum-number projection [1, 21], and for the energy it is a finite rodeo super-iteration [22] (whose infinite form gives $\sin x/x$). One block is a Dirichlet kernel,

$$\prod_{k=0}^{K-1} \cos(2^k E\tau) = \frac{\sin(2^K E\tau)}{2^K \sin(E\tau)}, \tag{3}$$

which equals 1 at $E = 0$. The identity is exact for every E (repeated use of $\sin 2x = 2\sin x\cos x$). The suppression bound is a separate step that uses the spectral window: for $E \in [\Delta, 1]$, $E\tau \in [\theta, \pi-\theta]$, so $\sin(E\tau) \ge \sin\theta$ and, with $|\sin(2^K E\tau)| \le 1$, the block is at most $\eta_b = 1/(2^K \sin\theta)$ in absolute value there. This bound relies on the scaling H_s, which places every relevant level in $[0, 1]$, and on the choice of τ; it does not hold outside $[\Delta, 1]$. With $r = \lceil \ln\eta / \ln\eta_b \rceil$ blocks, $T = r\tau(2^K - 1)$; choosing $2^K \sin\theta \approx e$ gives

$$T_{\text{dyadic}} \approx \frac{e\,\tau}{\sin\theta} \ln\frac{1}{\eta} \;\approx\; \frac{e}{\Delta} \ln\frac{1}{\eta}, \tag{4}$$

the optimal $O(\Delta^{-1}\log\eta^{-1})$ scaling [7] with an explicit constant and no optimization; Cohen and Oh reach the same conclusion for super-iterations on a generic gapped spectrum [22]. The pipeline searches K and r for the smallest T. Both closed-form schedules are certified by the inequalities just given (with $F_{\text{pass}} = \cos^m(w_0 t)$ and $1 - Lw_0$, respectively), which avoids interval arithmetic on filters with $L \sim 10^3$–10^5; the test suite checks them against a 2×10^6-point grid.

Optimized: times and phases chosen to maximize $F(0)^2$ subject to the certified constraint on η at fixed (T, m), over a grid of $x = T\Delta/\pi$ and $m \in \{4, 6, 8\}$, picking the pair with the smallest $T^2/F(0)^2$. This is SBC's recipe for energy filters [1] with a certified constraint in place of a grid objective; it uses the SLSQP designer `sbpipe/floor.py`. Nonzero phases trade success probability ($F(0) < 1$) for a shorter T.

For each schedule the leakage share $\epsilon_\ell/\epsilon \in \{0.5, 0.25, 0.1, 0.05\}$ is scanned and the plan with the smallest expected CNOT count per success is kept. An "auto" mode does the same over the dyadic and optimized schedules together.

C. Trotterization

Each controlled evolution $e^{-it_i H_s \otimes Z_a}$ is approximated by k_i steps of the first-order (Lie) or second-order (Strang) product formula over the non-identity terms; the identity term of H_s becomes a single ancilla R_z per pulse and is exact. For $B_a = \sum_{b>a} H_b$ the bounds of Childs *et al.* [2, Props. 15–16] are

$$\|S_1(t) - e^{-itH}\| \le \tfrac{t^2}{2}\alpha_1, \quad \alpha_1 = \textstyle\sum_a \|[B_a, H_a]\|, \tag{5}$$

$$\|S_2(t) - e^{-itH}\| \le t^3\beta_2, \quad \beta_2 = \tfrac{1}{12}\textstyle\sum_a \|[B_a,[B_a,H_a]]\| + \tfrac{1}{24}\textstyle\sum_a \|[H_a,[H_a,B_a]]\|, \tag{6}$$

so k steps over time t cost $\alpha_1 t^2/(2k)$ or $\beta_2 t^3/k^2$. The filter times t_i are in units of the scaled Hamiltonian, so the constants are evaluated for the terms of H_s, i.e. for coefficients c_a/W (the identity shift does not enter any commutator). Equivalently $\alpha_1(H_s) = \alpha_1(H)/W^2$ and $\beta_2(H_s) = \beta_2(H)/W^3$, with physical evolution time t_i/W; the code uses the first convention, and we checked that it agrees with the second to machine precision. Tensoring every term with Z_a leaves α_1 and β_2 unchanged because $Z_a^2 = 1$. For Pauli strings $[P_b, P_a] = 2P_bP_a$ when they anticommute and 0 otherwise, so each commutator is a short Pauli sum supported on a few qubits around term a; we evaluate its norm exactly on that support (dense for ≤ 8 qubits, Lanczos to a relative tolerance of 10^{-12} up to 12 with a 10^{-9} safety factor, triangle inequality beyond, which never occurred here). The Lanczos values are numerically exact but not certified upper bounds; replacing them by the triangle inequality would make every constant rigorous at a higher cost. The constants are maximized over the forward and reversed term order, so they hold for either convention. Every pulse evolves under the same $H_s \otimes Z_a$, so the per-pulse bounds $\alpha_1 t_i^2/(2k_i)$ and $\beta_2|t_i|^3/k_i^2$ share one constant; the phase ϕ_i enters only the ancilla rotation, and the bound is symmetric in the sign of t_i. With equal constants, the continuous relaxation of $\min \sum_i k_i$ subject to $\sum_i(\text{bound}_i) \le \epsilon_T$ is solved by $k_i \propto |t_i|$ for both orders (Lagrange). We take $k_i = \lceil |t_i|/\delta t \rceil$ with the common step δt of that solution. Rounding up keeps the bound, costs at most m steps more than the continuous optimum and need not be the integer optimum; the budget it frees goes to rotation synthesis. The certified times are untouched.

D. The certificate

Theorem 1. *Assume* (1) *and that the trial state has ground-space weight at least* γ. *Let the designed filter have certified* η *and* F_{pass}, *let* ϵ_U *bound the operator-norm error of the implementation (Trotter error* $\sum_i \epsilon_i$ *plus* $R\delta$ *for the* R *arbitrary rotations of the trial and filter circuits,*

each synthesized to accuracy δ), and let

$$\ell = \frac{(1-\gamma)\eta^2}{\gamma + (1-\gamma)\eta^2}, \qquad p_g = \gamma F_{\text{pass}}^2. \tag{7}$$

Let $|g\rangle = \Pi_0|\psi\rangle / \|\Pi_0|\psi\rangle\|$ be the normalized projection of the trial state onto the ground space, with Π_0 the ground-space projector; $|g\rangle$ is a pure state also when the ground space is degenerate. For a noiseless implementation the state $|\tilde{\psi}_{\text{out}}\rangle$ prepared conditioned on the ancilla outcomes 0 is pure, and

$$D \le \sqrt{\ell} + \frac{\epsilon_U}{\sqrt{p_g}}, \qquad 1 - \mathrm{Tr}(\Pi_0 \tilde{\rho}) \le 1 - |\langle g|\tilde{\psi}_{\text{out}}\rangle|^2 = D^2, \tag{8}$$

where D is the trace distance between $\tilde{\rho} = |\tilde{\psi}_{\text{out}}\rangle\langle\tilde{\psi}_{\text{out}}|$ and $|g\rangle$. Moreover $\langle H\rangle - E_0 \le (E_{\text{top}} - E_0)D^2$, and the success probability is at least $(\sqrt{p_g} - \epsilon_U)^2$. Below, $1-F$ denotes $1-|\langle g|\tilde{\psi}_{\text{out}}\rangle|^2$; for the nondegenerate ground state of the chains studied here it is the usual infidelity.

The proof (Appendix A) combines a leakage lemma for the exact filter with the observation that a vector error ϵ_U on an unnormalized vector of norm at least $\sqrt{p_g}$ moves its direction by at most $\epsilon_U/\sqrt{p_g}$. The pipeline sets the budget $\epsilon_U \le (\sqrt{\epsilon} - \sqrt{\ell})\sqrt{p_g}$, gives 90% of it to the product formula and the rest to rotation synthesis. If the trial state already satisfies $1 - \gamma \le \epsilon$, the pipeline also offers the empty filter ($\eta = 1$, no post-selection, $\ell = 1-\gamma$) and keeps whichever plan is cheaper.

E. Resource model

All-to-all connectivity and the Clifford+R_z gate set are assumed. Every Trotter factor of a pulse is controlled by the ancilla. For a Heisenberg bond (i, j) the three commuting terms form one exact three-qubit block,

$$e^{-ict(X_iX_j + Y_iY_j + Z_iZ_j)\otimes Z_a} = V^\dagger e^{-ict(Z_iZ_a + Z_jZ_a - Z_iZ_jZ_a)} V,$$
$$V = \mathsf{H}_i \,\mathrm{CNOT}_{ij}, \tag{9}$$

since the Bell-basis change V maps XX, YY, ZZ to $Z_i, -Z_iZ_j, Z_j$. The diagonal factor is a phase polynomial on the parities $i \oplus a$, $i \oplus j \oplus a$, $j \oplus a$, applied by $\mathrm{CNOT}_{ia}\, R_z\, \mathrm{CNOT}_{ja}\, R_z\, \mathrm{CNOT}_{ia}\, R_z\, \mathrm{CNOT}_{ja}$ on the ancilla wire [34]. One controlled bond thus costs 6 CNOTs and 3 R_z, against 12 CNOTs for three separate weight-3 Pauli rotations and 8 for qiskit's `PauliEvolutionGate` synthesis (both checked in the test suite, as is Eq. (9)); it is the controlled analogue of the 3-CNOT two-qubit gate [32, 33]. A numerical search over 300 four-CNOT layouts found no exact circuit. The block is the same unitary as its three term rotations, so the Trotterized circuit and its error are those of the term-wise product formula. The cheaper control by an anticommuting Pauli string, used by controlled-reversal rodeo circuits [10] and control-free QETU [9], is not available: no product of single-qubit Paulis anticommutes with all of XX, YY and ZZ on a bond. A Lie step costs one block per bond; k Strang steps cost $k(2B-2)+1$ blocks for B bonds, because the two middle half-steps and the outer half-steps of adjacent steps merge. Each pulse adds one ancilla R_z. An arbitrary rotation at accuracy δ costs $3\log_2(1/\delta)$ T gates, the leading term of Ross–Selinger synthesis [14]. The cost of one attempt is the trial circuit plus the filter, and the expected cost per success is the attempt cost divided by the certified lower bound $(\sqrt{p_g} - \epsilon_U)^2$ on the success probability of the implemented circuit from Theorem 1 (repeat-until-success; an early-abort protocol that measures after each pulse would be cheaper and is not modeled). The pipeline spends the whole budget $\epsilon_U = (\sqrt{\epsilon} - \sqrt{\ell})\sqrt{p_g}$, so this denominator is $p_g(1 - \sqrt{\epsilon} + \sqrt{\ell})^2 \ge (1 - \sqrt{\epsilon})^2 p_g$, i.e. at least $0.81\, p_g$ at $\epsilon = 10^{-2}$. Without a filter nothing is post-selected and every attempt succeeds. Alongside each T count we report the number of arbitrary rotations per attempt and their common synthesis accuracy δ.

III. MODEL, INPUTS AND VALIDATION

A. Model and front end

We study the open spin-$\frac{1}{2}$ chain

$$H = J_1 \sum_{i=1}^{N-1} \mathbf{S}_i \cdot \mathbf{S}_{i+1} + J_2 \sum_{i=1}^{N-2} \mathbf{S}_i \cdot \mathbf{S}_{i+2}, \tag{10}$$

$J_1 = 1$, at $J_2 = 0$ (Heisenberg), $J_2 = 0.2411$ (close to the critical point of the gapless-to-dimerized transition [16, 17]) and $J_2 = 0.4$ (dimerized), for $N = 8$ to 48. Its Pauli form has $B = N - 1$ bonds ($3B$ terms of weight 2) at $J_2 = 0$ and $B = 2N - 3$ otherwise. Terms are ordered by bond: nearest-neighbor bonds with even then odd left site, then next-nearest-neighbor bonds likewise.

The front end (`sbpipe/frontends/dmrg_quimb.py`) builds the MPO from the Pauli sum and runs two-site DMRG (quimb [18]) with bond dimensions up to 256 ($N \le 24$) or 128 ($N > 24$) and truncation cutoff 10^{-12}. E_1 is the lowest state of $H + \mu(S^z_{\text{tot}} - 1)^2$, $\mu = 2$, i.e. the lowest triplet, which we use as the global first excitation. The error bars $\delta E_0, \delta E_1$ are the energy change between the last two bond dimensions, floored at 10^{-10}; they are convergence heuristics, not bounds. Both DMRG energies are variational upper bounds, so a residual DMRG error in E_1 (or convergence of the penalty run to a higher state) biases the gap upward, which is the dangerous direction (Sec. V B); these error bars do not protect against that bias, and the certificate for $N > 14$ is conditional on the gap. The trial state is the variationally optimal MPS at fixed bond dimension $\chi \in \{1, 2, 4, 8\}$, and γ is its overlap with the reference MPS. This is an estimate of the overlap with the exact ground state, not a lower bound on it, so for $N > 14$ the certificate is equally conditional on γ. An MPS of bond dimension 2^b is exactly

TABLE I. DMRG inputs: ground energy, gap $E_1 - E_0$ and trial overlap γ of the optimal bond-dimension-χ MPS. Source: `results/inputs.json`.

N	J_2	E_0	gap	$\gamma_{\chi=1}$	γ_2	γ_4	γ_8
8	0	-3.374933	0.3927	0.134	0.775	0.999	1.0000
12	0	-5.142091	0.2809	0.060	0.591	0.994	1.0000
16	0	-6.911737	0.2193	0.028	0.475	0.984	0.9999
24	0	-10.453786	0.1530	0.006	0.328	0.954	0.9994
32	0	-13.997316	0.1177	0.001	0.234	0.913	0.9978
48	0	-21.085956	0.0809	0.000	0.125	0.817	0.9902
8	0.2411	-3.135001	0.3859	0.113	0.904	1.000	1.0000
12	0.2411	-4.738077	0.2720	0.046	0.806	1.000	1.0000
16	0.2411	-6.343316	0.2100	0.019	0.703	0.998	1.0000
24	0.2411	-9.556221	0.1443	0.003	0.517	0.989	0.9997
32	0.2411	-12.770432	0.1099	0.001	0.373	0.972	0.9989
48	0.2411	-19.200243	0.0744	0.000	0.191	0.920	0.9946
8	0.4	-3.025191	0.4037	0.036	0.976	1.000	1.0000
12	0.4	-4.545626	0.2956	0.003	0.945	1.000	1.0000
16	0.4	-6.066666	0.2371	0.000	0.911	0.999	1.0000
24	0.4	-9.109325	0.1748	0.000	0.839	0.995	0.9998
32	0.4	-12.152237	0.1418	0.000	0.768	0.990	0.9995
48	0.4	-18.238247	0.1075	0.000	0.640	0.976	0.9981

a staircase of $N - b$ unitaries on $b + 1$ qubits; we cost each with the quantum Shannon decomposition bound $\frac{23}{48}4^n - \frac{3}{2}2^n + \frac{4}{3}$ CNOTs (3, 20, 100 for $n = 2, 3, 4$) [15] and $4^n - 1$ rotations.

For $N \leq 14$ we compare with exact diagonalization (ED). DMRG reproduces E_0 to 2.3×10^{-11}, the first excited level of any symmetry to 2.4×10^{-11} (so the first excitation is the triplet in every ED case), and the trial overlaps to 4.0×10^{-8} (largest difference, for a $\chi = 1$ trial). The Weyl bound equals the exact top of the spectrum to 1.2×10^{-14}. Table I lists the inputs. The trial overlap decays roughly exponentially with N at fixed χ, because a fixed bond dimension captures a fixed amount of entanglement per bond while the truncation error accumulates over N bonds, so the filter has more to do at larger N.

B. Validation of the certificate ($N \leq 14$)

For $N \in \{8, 10, 12, 14\}$, the three couplings, $\epsilon \in \{10^{-2}, 10^{-3}\}$, the three schedules and both Trotter orders we plan a circuit for the $\chi = 2$ trial state and, when its size permits, simulate it exactly: the ancilla-0 block of the Trotterized circuit is $\frac{1}{2}(e^{-i\phi}S(t) + e^{i\phi}S(-t))$ per pulse, applied to the trial statevector and compared with the ED ground state. Of 144 plans, 96 were simulated (the others, mostly uniform or first-order plans with 10^4–10^8 steps, exceed our simulation budget), and the certificate held in all 96 (Fig. 2).

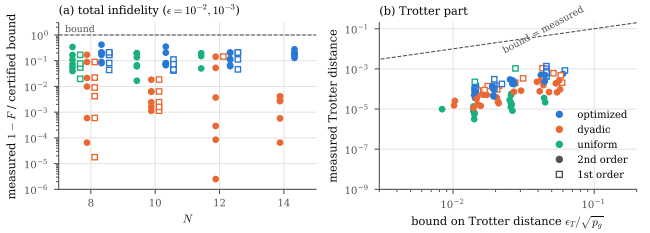


FIG. 2. All simulated plans. (a) Measured infidelity of the planned circuit divided by its certified bound; points below the dashed line respect the bound and are offset horizontally by schedule and order. (b) Measured distance between the Trotterized and the exact filtered state against its certified bound. Points below the dashed lines respect the bound. Filled circles: second order; open squares: first order.

The bound is conservative by a median factor 14 (range 2.4–4.0 $\times$ 10^5) in infidelity. Splitting it shows where: the leakage term ℓ overestimates the exact filter's infidelity by a median 4.3$\times$ (range 1.1–1.3 $\times$ 10^5). By schedule the medians are 1.4 (optimized), 4 (uniform) and 54 (dyadic): the optimized design pushes $|F|$ to ηF_{pass} across the band, whereas the closed-form dyadic bound $1/(2^K \sin\theta)$ is reached only near the band edge. More generally, η is a worst case over the whole stop band while the trial state's excited weight sits mostly low in the band. The Trotter term overestimates the measured Trotter distance by a median 185$\times$ (range 26–4.5 $\times$ 10^3): commutator bounds are worst-case over all states, while the filtered state lives in the low-energy subspace, where Trotter errors are much smaller [19, 31], and empirical Trotter errors are known to lie far below the bounds [30]. The measured success probability equals p_g to two digits in every case, because after filtering almost all of the surviving norm is ground-state weight.

Table II compares the schedules for one case. The uniform filter needs a total time an order of magnitude larger than the others, as its Δ^{-2} scaling predicts. The optimized filter is shortest; its nonzero phases lower p_g (0.58 against 0.81 here), which the cost criterion accounts for. Second order cuts the step count by one to two orders of magnitude at these sizes. Figure 3 shows the three filter functions.

How much of the remaining gap could a better bound recover? For $N \leq 12$ we bisect a common scale factor on the certified k_i until the simulated infidelity just meets ϵ. The certified step count is 8.7–39$\times$ (median 21) larger than this oracle value (Table III). The oracle uses the exact ground state and is not a usable rule; it measures the price of certification at second order.

TABLE II. Plans for $N = 12$, $J_2 = 0.2411$, $\chi = 2$ trial. $T_{\rm phys} = T/W$ in units of $1/J_1$; CNOT and T counts per attempt (filter) and expected per success (trial + filter, divided by the certified success probability $(\sqrt{p_g} - \epsilon_U)^2$). "rot.": arbitrary rotations per attempt (trial + filter), each synthesized to accuracy δ. "meas.": simulated infidelity of the planned circuit. Source: `results/validate.json`.

ϵ	schedule	order	m	$T_{\rm phys}$	p_g	steps	CNOT (filter)	CNOT/success	rot.	δ	T/success	meas. $1-F$
10^{-2}	uniform	1	437	164.2	0.81	4,136,205	5.2×10^{8}	7.2×10^{8}	2.6×10^{8}	2×10^{-11}	3.8×10^{10}	–
10^{-2}	uniform	2	371	139.4	0.81	17,437	4.2×10^{6}	5.5×10^{6}	2.1×10^{6}	1×10^{-9}	2.4×10^{8}	1.5×10^{-3}
10^{-2}	dyadic	1	10	23.3	0.81	77,308	9.7×10^{6}	1.4×10^{7}	4.9×10^{6}	1×10^{-9}	6.1×10^{8}	–
10^{-2}	dyadic	2	10	23.3	0.81	884	2.1×10^{5}	2.9×10^{5}	1.1×10^{5}	5×10^{-8}	1.1×10^{7}	8.5×10^{-7}
10^{-2}	optimized	1	6	8.7	0.58	13,384	1.7×10^{6}	3.2×10^{6}	8.4×10^{5}	5×10^{-9}	1.3×10^{8}	1.1×10^{-3}
10^{-2}	optimized	2	6	8.7	0.58	226	5.4×10^{4}	1.0×10^{5}	2.7×10^{4}	2×10^{-7}	3.5×10^{6}	1.1×10^{-3}
10^{-3}	uniform	1	745	279.9	0.81	27,817,555	3.5×10^{9}	4.5×10^{9}	1.8×10^{9}	1×10^{-12}	2.7×10^{11}	–
10^{-3}	uniform	2	658	247.2	0.81	55,930	1.3×10^{7}	1.7×10^{7}	6.7×10^{6}	2×10^{-10}	8.3×10^{8}	–
10^{-3}	dyadic	1	15	34.9	0.81	536,160	6.8×10^{7}	8.7×10^{7}	3.4×10^{7}	5×10^{-11}	4.5×10^{9}	–
10^{-3}	dyadic	2	15	34.9	0.81	2,835	6.8×10^{5}	8.8×10^{5}	3.4×10^{5}	5×10^{-9}	3.6×10^{7}	2.5×10^{-9}
10^{-3}	optimized	1	8	14.4	0.67	80,421	1.0×10^{7}	1.6×10^{7}	5.1×10^{6}	4×10^{-10}	7.4×10^{8}	–
10^{-3}	optimized	2	8	14.4	0.67	709	1.7×10^{5}	2.6×10^{5}	8.5×10^{4}	2×10^{-8}	1.0×10^{7}	6.2×10^{-5}

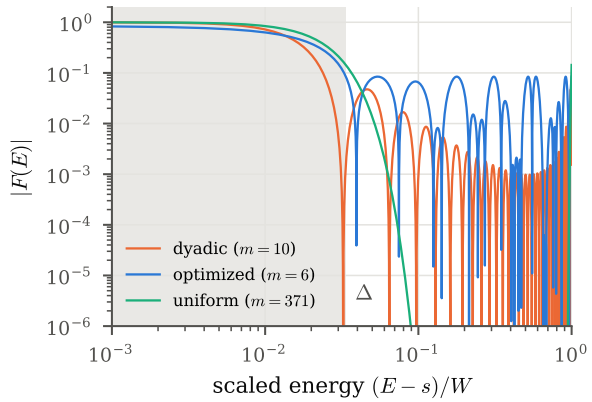


FIG. 3. $|F(E)|$ of the three schedules for $N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$, second order. Dotted lines: certified $\eta F_{\rm pass}$. Grey: the gap region. The horizontal axis is linear below Δ and logarithmic above.

IV. GAP AND OVERLAP AS INDEPENDENT INPUTS

The cost of the filter depends on two physical inputs, the gap above the ground state and the trial state's weight on it, and on two choices, the size N and the target ϵ. Along any natural family of chains these move together: at fixed J_2 the gap closes with N, and at fixed bond dimension the MPS overlap decays with N. Varying J_2 does not help, because near the dimerized point a larger gap and a better MPS come together. We therefore construct a family in which Δ and γ are set independently at every N, and measure the filter on it.

TABLE III. Certified over oracle step count (second order, $\chi = 2$). Columns: $N = 8, 10, 12$ for each J_2. Source: `results/oracle.json`.

		$J_2 = 0$			$J_2 = 0.2411$			$J_2 = 0.4$		
ϵ	schedule	8	10	12	8	10	12	8	10	12
1×10^{-2}	dyadic	13	22	30	24	24	34	18	15	28
1×10^{-2}	optimized	9	12	13	14	16	21	11	12	17
1×10^{-3}	dyadic	22	24	39	31	36	39	20	27	28
1×10^{-3}	optimized	12	17	21	24	22	27	14	16	17

A. Construction

Fixed gap. For each $N \in \{8, 10, 12, 14\}$ and target gap $g = E_1 - E_0 \in \{0.30, 0.35, \ldots, 0.55\}$ we find J_2 by bisection on exact diagonalization so that the singlet–triplet gap equals g to 10^{-9}. The gap closes with N at fixed J_2, so larger chains need larger J_2; pairs without a solution below the dimerized regime are left out (no J_2 gives $g \le 0.35$ at $N = 8$ or $g = 0.30$ at $N = 10$). The tuned couplings reach $J_2 = 0.68$, beyond the Majumdar–Ghosh point, where a singlet could in principle drop below the triplet. For all 26 tuned (N, J_2), including $N = 16$, the lowest excitation in the $S^z = 0$ block, which contains a member of every multiplet, coincides with the lowest $S^z = 1$ state to 10^{-7}, so the first excitation is the triplet and g is the true gap (`results/gapov_e1check.json`).

Fixed overlap with a real circuit. The trial is the variational $\chi = 16$ MPS, prepared by its staircase circuit, followed by $R_y(\theta) = e^{-i\theta Y/2}$ on every even site. A uniform rotation of all sites would leave the overlap unchanged,

because the ground state is a singlet; rotating one sublattice breaks that symmetry and lowers γ smoothly. θ is set by bisection so that γ hits the target to 10^{-12}. The trial cost is the MPS circuit plus $N/2$ single-qubit rotations at every γ, so the trial circuit does not change along an overlap scan.

Grid and measurements. Study A fixes $g \in \{0.40, 0.50\}$ and scans $\gamma \in \{0.03, 0.1, 0.3, 0.5, 0.7, 0.9\}$; study B fixes $\gamma \in \{0.1, 0.5\}$ and scans g over 0.30–0.55; both at $\epsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, 222 cases in all. Each case is run with both SBC schedules, dyadic and optimized, and with QETU for comparison (Sec. VII), and for each we record two costs. *Measured*, dyadic: the exact filter is applied to the trial state for each ladder (K, r); for the two shortest ladders that reach ϵ, the coarsest common Strang step at which the full Trotterized circuit still reaches ϵ is found by bisection on the state vector, and the cheaper is kept. *Measured*, optimized: SBC's design (Sec. II) for the target leakage ϵ with the true gap and overlap, then the same step search. The numerical design occasionally converges to a filter that barely passes the ground state, $f(E_0)^2 \sim 10^{-3}$, and with it a success probability far below γ; wherever $p < \gamma/2$ the design is repeated over four optimizer seeds and two design targets $(\epsilon, \epsilon/2)$ and the cheapest passing circuit is kept (21 of 286 points). *Certified*: Theorem 1 with exact energies (error bars 10^{-9}), the Weyl bound and the true γ. Both are reported per shot (one run of the circuit) and per success (divided by the success probability p), and, unless stated otherwise, for the filter alone, without the trial circuit.

B. The gap sets the circuit, the overlap sets the repetitions

Figures 4 and 5 show CNOTs, R_z rotations and evolution time per shot, and CNOTs per success, along both scans. The two inputs act on different resources.

Per shot, the overlap enters only logarithmically. At fixed gap, lowering γ thirtyfold (0.9 to 0.03) raises the measured CNOTs per shot of the dyadic filter by only 1.2–3.0× over all N and ϵ, and its evolution time by 1.5–3.0×. The filter must suppress the excited weight to $\eta^2 = \gamma\epsilon/(1-\gamma)$; a worse trial raises $(1-\gamma)/\gamma$ by a factor of 290 here, which adds only $\ln 290 \approx 5.7$ to $L = \ln[(1-\gamma)/(\gamma\epsilon)]$, already 2–13 across these cases. The dyadic filter grows in whole pulses, so a worse trial often needs no extra pulse at all; its per-shot cost moves in steps (Fig. 4, top).

Per shot, the gap enters as a power. At fixed overlap, closing the gap by 1.8× (0.55 to 0.30) raises the CNOTs per shot by 2.7–4.3× and the evolution time by 1.5–2.8× (Fig. 5). The time grows about as $1/\Delta$, the cost of resolving a smaller gap.

Per success, the overlap returns as $1/\gamma$. Every run succeeds with probability $p \approx \gamma f(E_0)^2$, and for the dyadic filter $f(E_0) = 1$, so the expected cost per success carries the factor $1/\gamma$ (last columns of Figs. 4 and 5). This is the only place where the overlap enters as a power. The practical reading is direct: the gap decides the depth of the coherent circuit, and with it whether the filter fits within a coherence or error budget at all; the overlap decides how many times that circuit has to be run.

R_z counts follow CNOT counts exactly. Every controlled bond block has 3 R_z and 6 CNOTs, and each pulse adds one ancilla rotation, so $R_z/\mathrm{CNOT} = 0.50$ in every case. Every statement about CNOTs below therefore holds for the arbitrary rotations, and so for the T count of a fault-tolerant implementation up to the synthesis factor $3\log_2(1/\delta)$.

C. Mechanism: filter length, Trotter step and gates per step

The per-shot cost factorizes as

$$\text{CNOTs per shot} = \frac{T}{\delta t} \times c_{\text{step}}(N), \tag{11}$$

and each factor answers to different inputs (Fig. 6).

Filter length. The physical evolution time T (in $1/J_1$) depends on Δ and L only. Fitting all cases to $a\, N^{\alpha} \Delta^{-\beta} L^{\lambda} \epsilon^{-e}$, the measured dyadic filter has $T \propto \Delta^{-1.19} L^{0.77}$ with $N^{-0.24}$, and the certified one $T \propto \Delta^{-1.10} L^{0.84}$ with $N^{0.04}$: at a fixed gap in absolute units, the filter's length does not depend on the size of the system. The certified length is a median 2.0× the measured one (range 1.2–3): the certificate sizes the filter nearly right.

Trotter step. The measured dyadic filter tolerates very coarse steps: median $\delta t = 1.25$, 0.56, 0.30 (in $1/J_1$) at $\epsilon = 10^{-2}$, 10^{-3}, 10^{-4}, growing with the gap. Its pulses are long and smooth, and the filtered state lives at the bottom of the spectrum, where Trotter errors are much smaller than their worst case [19]. The certified step must cover the worst case over the whole evolution: median $\delta t = 0.017$, 0.009, 0.005, about 60–75 times finer, and shrinking as the total time grows. As a result the certified step count grows as $L^{2.03}$ against $L^{0.71}$ measured, and is a median 89.4× (range 26.2–339) the measured one. Almost all of the certificate's overhead is in the Trotter bound, not in the filter.

Gates per step. One Strang step costs a median $20\,N$ CNOTs for the two-neighbor chain ($6(2B-2)$ with $B = 2N-3$ bonds). The measured dyadic step count barely grows with N (exponent 0.08), so its CNOTs grow as $N^{1.30}$; the certified step count grows as $N^{0.85}$ through the commutator constant and the scaled time, giving $N^{2.08}$.

D. Cost laws

Fitting the expected CNOTs per success (filter and trial circuit, as a user would pay them) to

$$C \approx a\, N^{\alpha}\, \Delta^{-\beta}\, L^{\lambda}\, \gamma^{-g}, \qquad L = \ln\frac{1-\gamma}{\gamma\epsilon}, \tag{12}$$

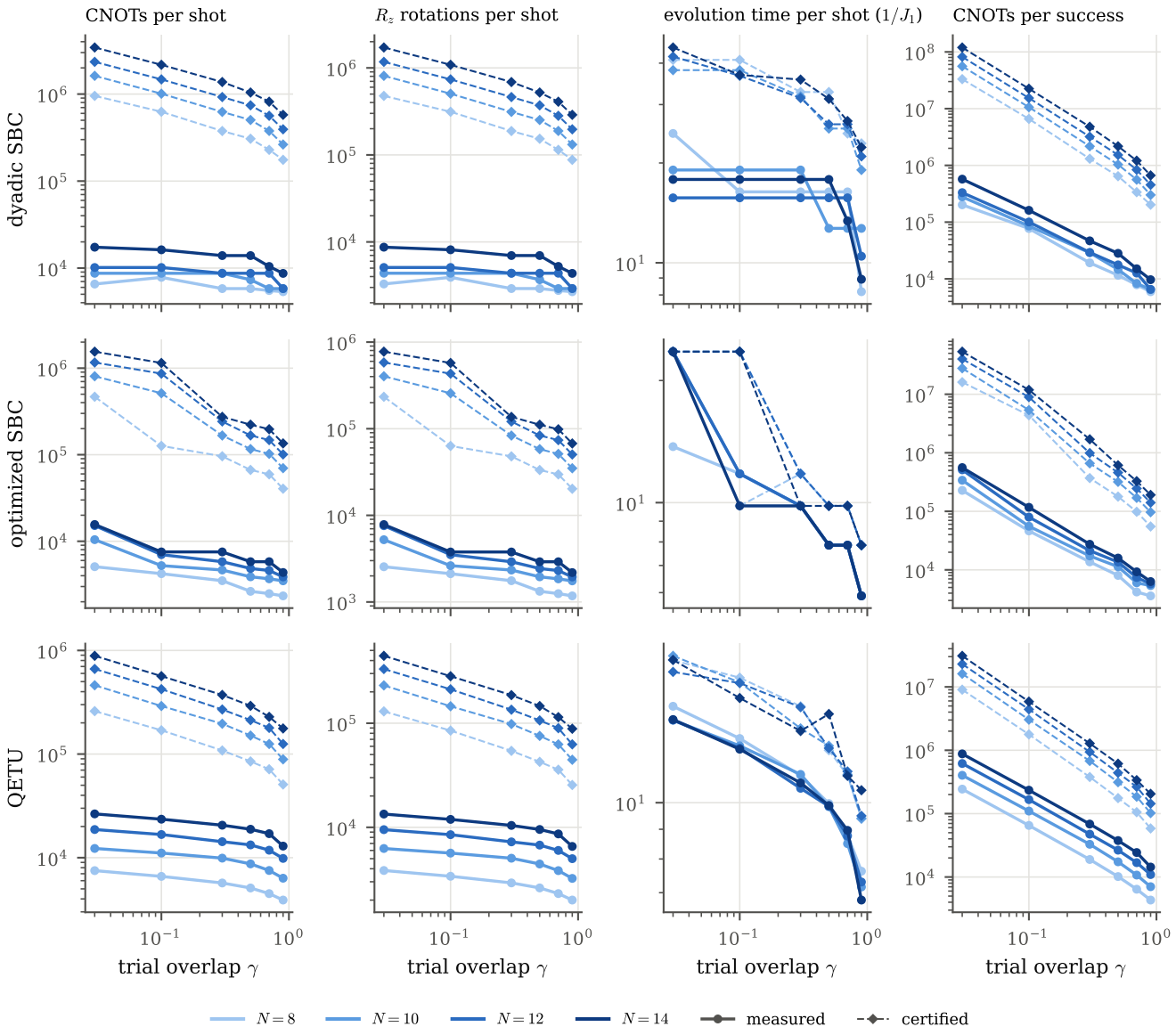


FIG. 4. Study A: filter-only resources at a gap held at $g = 0.40$ for every N, $\epsilon = 10^{-3}$, against the trial overlap. Rows: dyadic SBC, optimized SBC, QETU (Sec. VII). Solid: measured by simulation; dashed: certified. Columns 1–3 per shot, column 4 per success. Source: `results/gapov.jsonl`, `results/gapov_filter.json`.

over all 222 cases gives Table IV. For the dyadic filter,

$$C_{\text{meas}} \approx 10.1\, N^{1.50} \Delta^{-1.15} L^{1.41} \gamma^{-0.78}, \tag{13}$$

$$C_{\text{cert}} \approx 15\, N^{2.07} \Delta^{-1.27} L^{2.43} \gamma^{-0.98}, \tag{14}$$

within a typical factor of 1.23 and 1.22 respectively. The certified law has the $1/\gamma$ of the repetitions and an $L^{2.4}$ from the Trotter bound; the measured law grows more slowly in both N and L. The measured dyadic γ exponent is below one because its success probability ranges from 0.76 to 1.24 times γ: at the coarse steps it tolerates, each Trotterized pulse is slightly off-diagonal in the eigenbasis and moves weight into or out of the ground state. Part of its γ dependence is therefore absorbed into L. Across all cases the certified cost lies 14–258× (median 62) above the measured one. These exponents summarize $N = 8$–14 and $g = 0.30$–0.55; they describe the data and are not asymptotic laws.

E. Dyadic versus optimized SBC

The two SBC schedules reach the same physics by different routes. The dyadic ladder has zero phases, so it passes the ground state with $f(E_0) = 1$ and its length grows in whole pulses. The optimized design (times and phases from SBC's recipe with a certified suppression constraint) is shorter, at the price of $f(E_0) < 1$: its

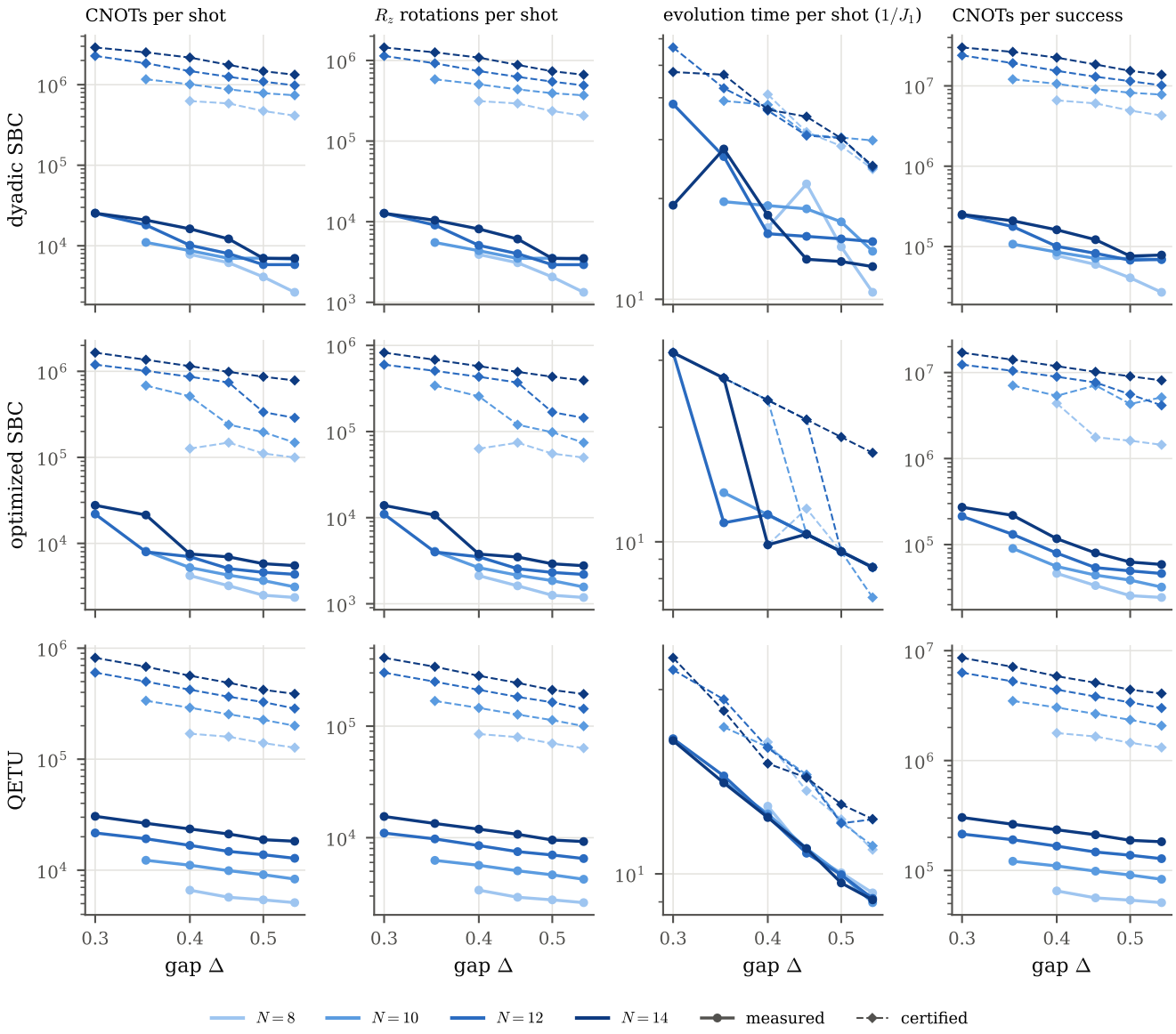


FIG. 5. Study B: as Fig. 4 at an overlap held at $\gamma = 0.10$ for every N, against the gap (held fixed across N by tuning J_2).

measured success probability is a median 0.88 times γ (as low as 0.52). Measured per shot, the optimized filter uses a median 0.58 times the dyadic evolution time and 0.62 times its Trotter steps; its tolerated step is the same, median $\delta t = 1.24$ at $\epsilon = 10^{-2}$ and 0.30 at 10^{-4}. Per success, after the lower success probability is paid, it is still a median 0.89× the dyadic cost (range 0.58–1.55) and the cheaper one in 169 of 222 cases. Certified, the advantage is larger, a median 0.35× (cheaper in 221 of 222), because the optimized filter's leakage bound is nearly tight while the closed-form dyadic bound is loose (Sec. III B), and the shorter filter needs a looser Trotter budget; its certified cost lies 5–185× (median 26) above the measured one, against 62× for dyadic.

Counting the cheapest of the three filters case by case, the optimized SBC filter wins 51 of 74 cases at $\epsilon = 10^{-2}$ (dyadic the other 23), 57 at 10^{-3} (dyadic 16, QETU 1), and 32 at 10^{-4}, where QETU takes 37 and dyadic 5. The cost laws of Table IV make the trade-off explicit: the optimized measured cost grows more steeply with the gap ($\Delta^{-1.36}$ against $\Delta^{-1.15}$), because its design is matched to the gap, and less steeply with L. The price of the optimized schedule is its design: a nonconvex optimization of about 40 s per case that converged to a filter with $f(E_0)^2 \sim 10^{-3}$ in 21 of 286 measured points and had to be repeated with other seeds; the dyadic ladder needs no design and never fails.

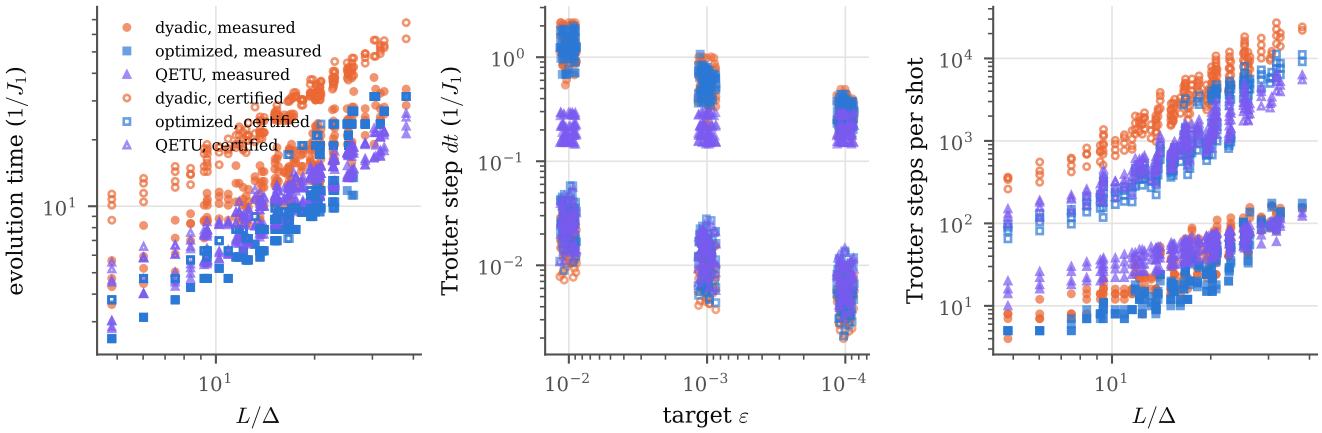


FIG. 6. All 222 cases. (a) Evolution time against L/Δ, $L = \ln[(1-\gamma)/(\gamma\epsilon)]$. (b) Trotter step against ϵ (jittered horizontally). (c) Trotter steps per shot against L/Δ. Filled: measured; open: certified. Orange circles: dyadic SBC; blue squares: optimized SBC; purple triangles: QETU.

TABLE IV. Fits of the expected CNOTs per success to Eq. (12) over all 222 gap/overlap cases. "rms"/"max": typical and worst factor between a case and its fit. Source: `results/gapov.jsonl`.

	cases	a	α	β	λ	g	rms	max
dyadic, measured	222	1.0×10^1	1.50	1.15	1.41	0.78	1.23	1.9
optimized, measured	222	9.8	1.50	1.36	1.24	0.83	1.30	2.3
QETU, measured	222	1.3×10^1	2.09	0.74	0.68	0.99	1.04	1.2
dyadic, certified	222	1.5×10^1	2.07	1.27	2.43	0.98	1.22	2.2
optimized, certified	222	1.9	2.27	1.30	2.74	0.97	1.45	3.2
QETU, certified	222	9.5	2.03	1.11	2.17	0.99	1.22	2.0

F. Cost maps

Two maps make the interplay visible. At fixed N and $\epsilon = 10^{-2}$ (Fig. 7), the contours of the dyadic cost per success run nearly parallel to the gap axis for $g \gtrsim 0.40$: there the overlap alone sets the cost, from 2–6 $\times 10^3$ CNOTs at $\gamma = 0.9$ to 1–4 $\times 10^5$ at $\gamma = 0.03$; below $g \approx 0.40$ the gap starts to matter, and $g = 0.30$ costs 1.3–2.7 times $g = 0.40$ at the same overlap. The optimized filter is cheaper over most of the map, and its contours tilt toward the gap axis over the whole range, as its steeper gap exponent implies. Per shot (not shown) the contours of both turn by ninety degrees and the gap alone sets the circuit. At fixed gap and overlap (Fig. 8), the measured cost per success grows 7–21× (dyadic) from $\epsilon = 10^{-2}$ to 10^{-5} and 1.8–4.6× from $N = 8$ to 14; the contours tilt toward the ϵ axis, so over this range the target matters more than the size. The optimized filter's lead is largest at loose targets and at $\gamma = 0.5$, and shrinks toward $\epsilon = 10^{-5}$.

V. RESOURCES AND REGIMES OF ADVANTAGE

The controlled study fixes the inputs; here they take the values that DMRG supplies for natural chains, at sizes beyond exact simulation, and we ask when filtering beats the alternative of compiling a better MPS.

A. Resources up to $N = 48$

Without simulation the pipeline only needs DMRG, so we plan for N up to 48 with the same $\chi = 2$ trial family (Fig. 9, Table V). Over all sizes, couplings and targets the second-order plan is 17–666× (median 93) cheaper than the first-order plan of the same schedule. The optimized schedule is 0.7–3.7× (median 2.9) cheaper than the dyadic one at second order, and more expensive in 1 of 29 comparable cases. Its numerical design, however, failed in 14 of 144 plans, all at $N \geq 32$ where $\gamma \lesssim 0.4$ or $\epsilon = 10^{-3}$ demand a suppression η the SLSQP search with $F(0)^2 \geq 10^{-3}$ did not reach; the dyadic schedule is feasible everywhere, which is why the pipeline offers an automatic choice between the two. For $N = 16$–48 the expected CNOT count of the dyadic second-order plan grows as $N^{4.9}$, $N^{5.0}$ and $N^{3.5}$ for $J_2 = 0, 0.2411, 0.4$, and its step count as $N^{2.6}$, $N^{2.7}$, $N^{2.1}$. These exponents combine the shrinking gap ($T \propto 1/\Delta$, and $\Delta = g/W$ with $W \propto N$), the growing commutator constant, the number of terms per step and the decaying overlap (through the success probability); they are fits over $N = 16$–48, not asymptotic statements. At $N = 48$, $J_2 = 0.2411$ the best $\chi = 2$ plan (dyadic) needs 2.7×10^8 CNOTs per success, which Sec. V C reduces by choosing a better trial.

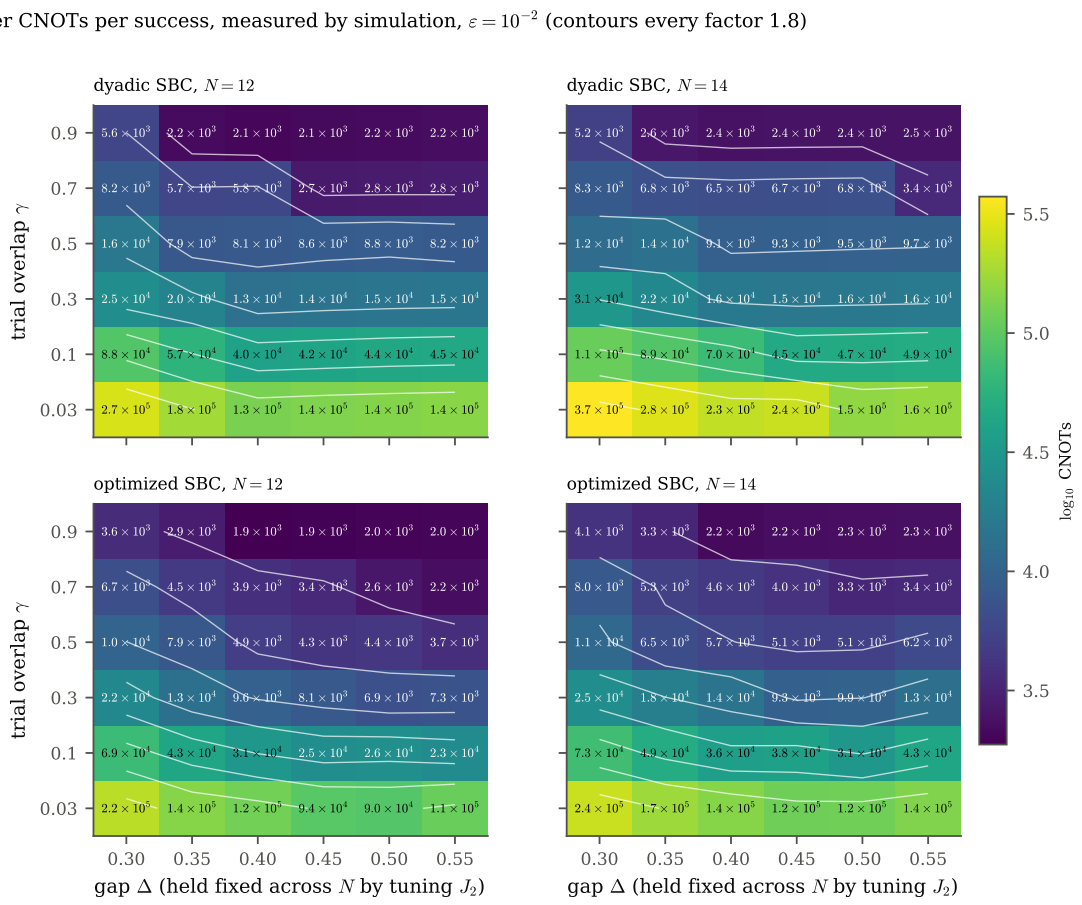


FIG. 7. Filter CNOTs per success over gap and overlap at $\epsilon = 10^{-2}$, measured, for the dyadic (top) and optimized (bottom) SBC filters at $N = 12$ and 14. White contours every factor 1.8. Source: `results/gapov.jsonl`, `results/gapov_grid.jsonl`.

B. What if the energy estimates are wrong?

The certificate is only as good as its inputs, so we perturb them for $N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$ (optimized, second order) and simulate the resulting circuit on the true trial state (Fig. 10).

Gap. A gap assumed larger than the true one moves the stop-band edge above the first excitation, which then sits in the filter's transition region. The target was missed in 3 cases, $f \geq 1.5$, with infidelity up to 5.1×10^{-2} at $f = 3$; at $f = 1.1$ and 1.25 the looseness of the certificate still covered the error. Underestimating the gap is safe and cheap: even $f = 0.5$ costs only 2.9×. Since DMRG gives an *upper* bound on E_1, an explicit margin δE_1 is the natural safeguard.

Ground energy. An error bar δE_0 widens the pass band and lowers F_{pass}; the cost stays flat until δE_0 is a sizeable fraction of the gap (13.5× at $\delta E_0 = 0.1$ gap). DMRG energies are far more accurate than this, so E_0 is never the limiting input here.

Overlap. Underestimating γ is safe (a stricter η); the planned cost rises 9.2× at $f = 0.25$, mostly because the planned success probability $p_g = \gamma F_{\text{pass}}^2$ uses the pessimistic γ, while the actual success rate does not change. Overestimating it by 2% did not break the target in this instance (0 failures), because the leakage term is itself loose.

All instances. To see whether this one instance is typical we repeated the gap and overlap perturbations for every $N \leq 14$, all three couplings and the dyadic, optimized and QETU schedules (second order, $\epsilon = 10^{-2}$; Table VI). With exact inputs all 36 plans meet the target. An overestimate of the gap by 10% already breaks it in 2 of 36 plans (marginally, worst 1.1×10^{-2}), by 25% in 10 and by 50% in 16. The closed-form dyadic schedule is the most tolerant (0 misses at 25%), because its bound is the loosest; the optimized and Chebyshev filters, whose leakage terms are nearly tight, have no slack to absorb a wrong gap. Overestimating γ broke the target in 12 of 72 plans, all of them through the same mechanism: the inflated γ crossed $1 - \epsilon$, so the pipeline dropped the filter altogether (12 of 12). A margin on both inputs is therefore not optional: δE_1 should be a sizeable fraction of the gap when E_1 is not certified, and the empty filter should be accepted only on a certified (for example Eckart) lower bound on γ.

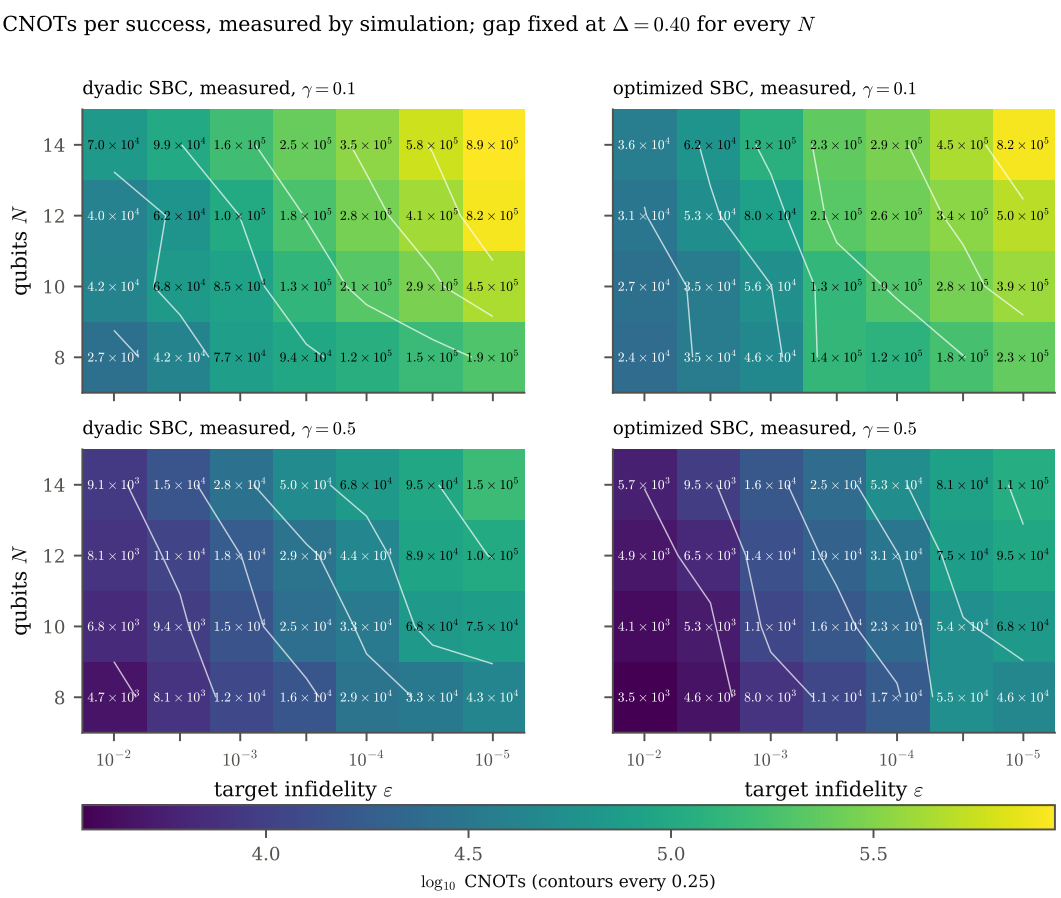


FIG. 8. Filter CNOTs per success over N and ϵ at a gap held at $g = 0.40$, measured, for the dyadic (left) and optimized (right) SBC filters at $\gamma = 0.1$ (top) and 0.5 (bottom). Source: `results/gapov*.jsonl`.

C. Choosing the ansatz

The trial bond dimension trades circuit depth for overlap. Figure 11 and Table VII show the expected total cost for $\chi = 1$ to 8 (the pipeline choosing the cheaper of the dyadic and optimized schedules). A product state ($\chi = 1$) has an exponentially small overlap and is by far the most expensive start (10^7–10^{23} CNOTs). Going to $\chi = 2$ and $\chi = 4$ lowers the cost by orders of magnitude: at $N = 48$, $J_2 = 0.2411$ the $\chi = 4$ plan needs 4.8×10^6 CNOTs and 2.1×10^8 T gates per success. At $\chi = 8$ the trial state alone satisfies $1 - \gamma \leq 10^{-2}$ in 9 of 9 cases ($\gamma = 0.9946$ at $N = 48$, $J_2 = 0.2411$), and the pipeline returns the empty filter: 4,500 CNOTs. For this gapless-to-weakly-gapped one-dimensional chain at $\epsilon = 10^{-2}$, directly compiling a better MPS is therefore much cheaper than filtering a worse one. The filter becomes the better tool when the target is tighter than what a compilable MPS reaches, when a certificate on the output is required, or when the classical method cannot provide a compact high-overlap trial circuit, as for two-dimensional lattices.

D. Measured crossover

The comparison above is certified on both sides only for the MPS. To compare like with like we measure the filtered route too: for chains of $N = 12$–16 at $J_2 = 0$, 0.2411 and 0.4, the exact spectrum of each Hamiltonian is computed per S^z block, every MPS trial with $\chi = 2, \ldots, 64$ is decomposed on it, and for each target $\epsilon \in [10^{-5}, 10^{-2}]$ the cheapest route is found: the smallest MPS that meets $1 - \gamma \leq \epsilon$ alone, against the cheapest smaller MPS followed by a measured dyadic or optimized SBC filter (shortest exact ladder or design, then the coarsest passing Strang step), with cost (trial + filter)/p.

Figure 12 shows the result on nine chains. Compiling the MPS directly is the cheaper route in most cases: the filtered route wins in 18 of the 63 (chain, ϵ) targets, by 1.1–3.2×. The wins share a single mechanism. Because MPS compilation cost jumps with χ (here by a factor of about four from $\chi = 4$ to 8 and from 8 to 16), a target lying just below what a given χ reaches forces the MPS-alone route up to the next bond dimension, while the trial at the lower χ misses the target only slightly. All 18

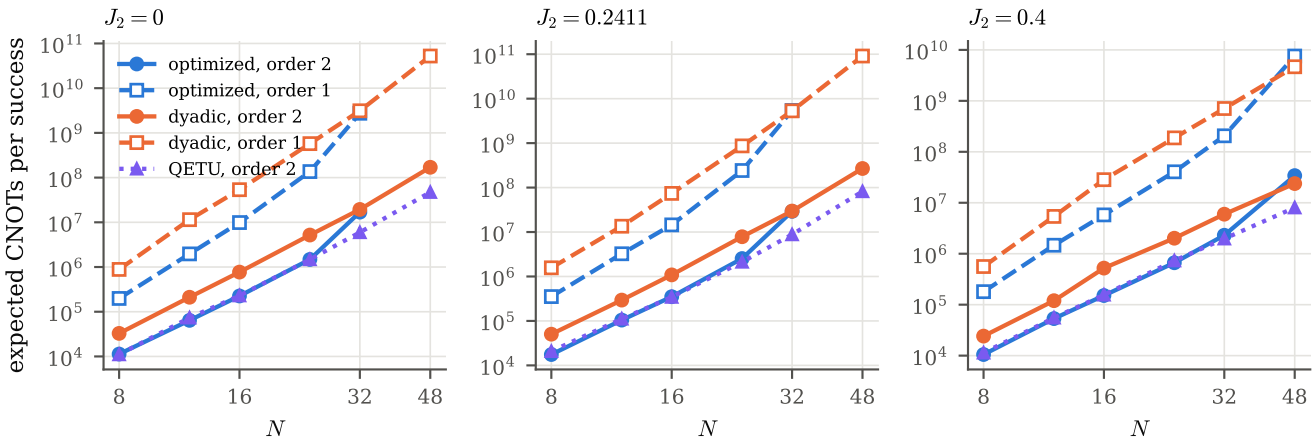


FIG. 9. Expected CNOTs per successful preparation versus N (trial $\chi = 2$, $\epsilon = 10^{-2}$), including the trial circuit and the repetitions (certified success probability). Solid: second order; dashed: first order. Missing optimized points: no feasible design (see text). Dotted triangles: QETU, second order. Sources: `results/scaling.json`, `results/ext_qetu_scaling.json`.

TABLE V. Cheapest second-order plan ("sch.": dyadic or optimized) for $\epsilon = 10^{-2}$, trial $\chi = 2$. Δ is the scaled gap, $T_{\text{phys}} = T/W$ in $1/J_1$. Steps for first (1st) and second (2nd) order with the same schedule; CNOT and T counts for second order per attempt (att.) and per success (succ.), divided by the certified success probability; "rot.": arbitrary rotations per attempt, synthesized to accuracy δ. Source: `results/scaling.json`.

N	J_2	γ	Δ	sch.	T_{phys}	p_g	steps (1st)	steps (2nd)	CNOT/att.	CNOT/succ.	rot.	δ	T/succ.
16	0	0.475	0.0206	opt.	14.3	0.38	38,020	464	7.8×10^4	2.3×10^5	3.9×10^4	9×10^{-8}	8.0×10^6
16	0.2411	0.703	0.0192	opt.	15.0	0.58	41,572	521	1.8×10^5	3.5×10^5	8.8×10^4	7×10^{-8}	1.3×10^7
16	0.4	0.911	0.0211	opt.	6.6	0.63	19,266	264	8.9×10^4	1.5×10^5	4.5×10^4	6×10^{-8}	5.4×10^6
24	0	0.328	0.0094	opt.	20.5	0.27	251,160	1,422	3.8×10^5	1.5×10^6	1.9×10^5	8×10^{-9}	6.0×10^7
24	0.2411	0.517	0.0087	opt.	16.3	0.29	248,256	1,329	7.0×10^5	2.5×10^6	3.5×10^5	5×10^{-9}	1.1×10^8
24	0.4	0.839	0.0102	opt.	10.8	0.59	87,533	700	3.7×10^5	6.6×10^5	1.9×10^5	1×10^{-8}	2.6×10^7
32	0	0.234	0.0054	opt.	80.0	0.23	3,103,919	9,847	3.5×10^6	1.7×10^7	1.8×10^6	1×10^{-9}	7.4×10^8
32	0.2411	0.373	0.0049	opt.	85.8	0.37	4,979,405	13,617	9.8×10^6	2.9×10^7	4.9×10^6	6×10^{-10}	1.3×10^9
32	0.4	0.768	0.0062	opt.	13.3	0.41	221,517	1,228	8.8×10^5	2.3×10^6	4.4×10^5	5×10^{-9}	9.6×10^7
48	0	0.125	0.0025	dya.	146.0	0.13	20,734,266	34,380	1.9×10^7	1.7×10^8	9.5×10^6	2×10^{-10}	8.2×10^9
48	0.2411	0.191	0.0022	dya.	142.5	0.19	27,888,279	41,586	4.6×10^7	2.7×10^8	2.3×10^7	1×10^{-10}	1.3×10^{10}
48	0.4	0.640	0.0031	dya.	69.3	0.64	4,816,485	12,414	1.4×10^7	2.4×10^7	6.9×10^6	6×10^{-10}	1.1×10^9

wins filter such a trial, with $(1-\gamma)/\epsilon \leq 8.8$ (13 of them below 3), using a dyadic ladder of one to four pulses ($K \leq 3$, $r \leq 2$, 1–21 Strang steps) that succeeds with $p \geq 0.988$. Of the 31 filtered routes with $(1-\gamma)/\epsilon \leq 8.8$, 18 beat the MPS alone; all 90 with a larger ratio lose, by a median factor of 20. In this regime the filter acts as a last-mile polisher: it removes the residual infidelity of a nearly sufficient trial, but recovering a larger factor is cheaper by doubling χ. The location of the wins in ϵ therefore depends on the system through where each χ lands, not on N or J_2 in any simple way, and an MPS with intermediate bond dimension (not tried here) would compete with some of them. The optimized design never wins. A single dyadic pulse lasts only $\pi/(1+\Delta)$ and keeps $f(E_0) = 1$; it does not resolve the gap, but it removes 17–68% of the residual weight, which suffices for a trial this close to the target. The optimized design certifies suppression on the whole band above Δ and so needs $T \gtrsim \pi/\Delta$: rerunning the competitive cases ($(1-\gamma)/\epsilon \leq 20$; 32 of 40 completed) with one- to three-pulse designs and the design target loosened up to 10ϵ lowered its cost by up to 3× but left $T = 35$–66 against 3–21 for the winning dyadic ladders, and it beat the dyadic route in 3 of the 23 cases with $(1-\gamma)/\epsilon \leq 8.8$ and the MPS alone in none (`results/emp_actual1d_opt.jsonl`). QETU beats the MPS alone in 11 of the targets but is never the cheapest route. This matches the observation for rodeo filters that a ratio-2 time series is optimal when the total time is

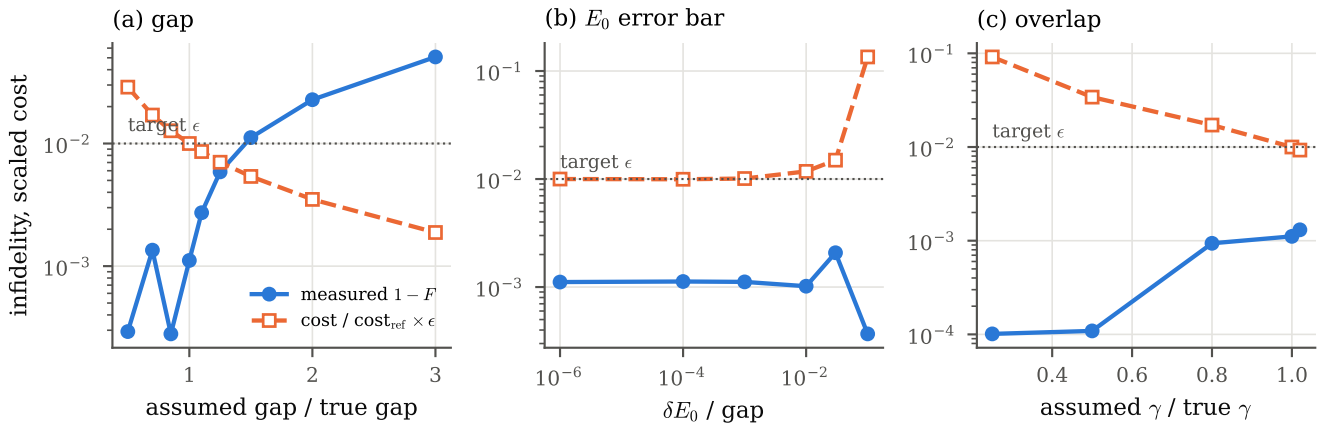


FIG. 10. Sensitivity to the inputs ($N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$). Circles: measured infidelity of the planned circuit on the true trial state. Squares: expected CNOT cost relative to the plan with exact inputs, multiplied by ϵ to share the axis. (a) Assumed gap $f\times$ the true gap; (b) error bar on E_0; (c) assumed overlap $f\times$ the true γ. Source: `results/sensitivity.json`.

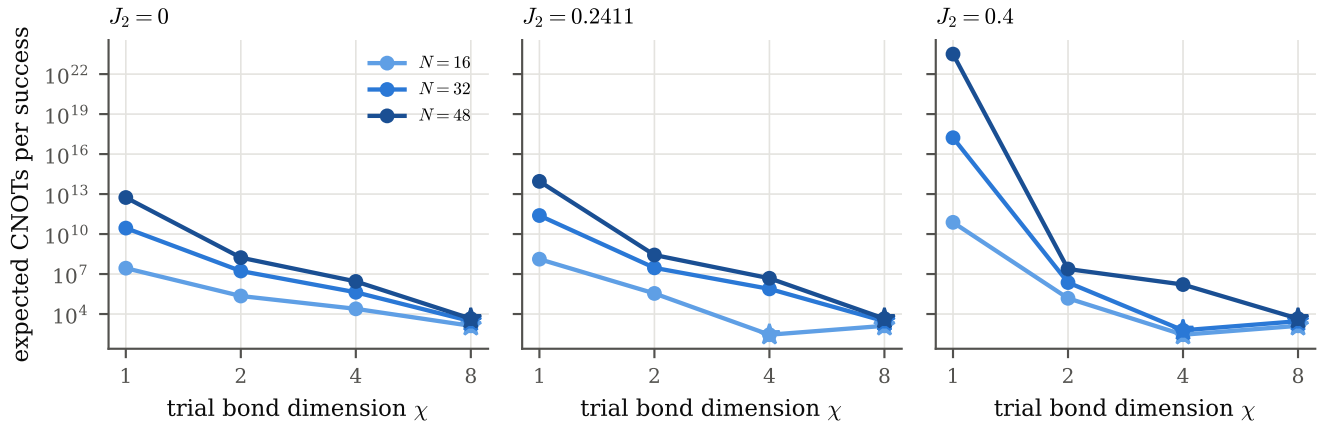


FIG. 11. Expected CNOTs per success versus trial bond dimension ($\epsilon = 10^{-2}$, second order, cheaper of dyadic and optimized). Stars: the trial state alone satisfies $1 - \gamma \le \epsilon$ and no filter is used. Source: `results/ansatz.json`.

short and longer series favor other designs [23]. Because the bond dimension a chain needs for fixed ϵ grows only slowly with N, we expect this picture to hold for longer chains; on cylinders it reverses (Sec. VI).

VI. TWO DIMENSIONS: WHERE THE FILTER PAYS

The pipeline is generic in H, so we ran the measured crossover of Sec. V D on the J_1–J_2 model on 4×3 and 4×4 cylinders (periodic around the four-site circumference, open along L_x) at $J_2 = 0$ and 0.3, with snake-path MPS trials up to $\chi = 64$ and exact spectra (Fig. 13). Cylinders are much harder for a snake MPS than chains. At 4×4, $J_2 = 0$, the $\chi = 16$ trial still misses $\epsilon = 10^{-2}$ ($1-\gamma = 1.1\times10^{-2}$), $\chi = 32$ stalls at 7×10^{-4}, and no $\chi \le 64$ trial reaches 10^{-5} ($1-\gamma \approx 2\times10^{-5}$ at $\chi = 64$), while the compile cost grows about fourfold per doubling of χ (5 328, 20 548 and 76 600 CNOTs for $\chi = 16$, 32, 64). On 4×3, $\chi = 64$ is exact, but only at 45 960 CNOTs.

The result is the opposite of the chain. The filtered route is cheaper at 27 of the 28 cylinder targets from $\epsilon = 10^{-2}$ to 10^{-5}, by 1.1–8.8× where both routes exist, and it is the only route at 10^{-5} on 4×4. The exception is $\epsilon = 10^{-2}$ on 4×4, $J_2 = 0.3$, where the $\chi = 16$ MPS alone already reaches 8.3×10^{-3}. Every win uses the dyadic ladder, on a $\chi = 16$ trial in 24 of the 27 (otherwise $\chi = 4$, 8 or 32), with $p \ge 0.77$ and 1–46 Strang steps. Unlike the chain, the filter here recovers large factors: the trials miss the target by $(1-\gamma)/\epsilon$ up to 10^3, and the advantage grows toward tight targets (up to 8.8× at 4×4, $J_2 = 0$, $\epsilon = 3\times10^{-4}$, where $\chi = 16$ plus 10 Strang steps replaces $\chi = 64$). Two properties of the filter carry this: its depth grows only with $\ln(1/\epsilon)$ at a fixed gap, while the MPS needed to reach ϵ alone grows

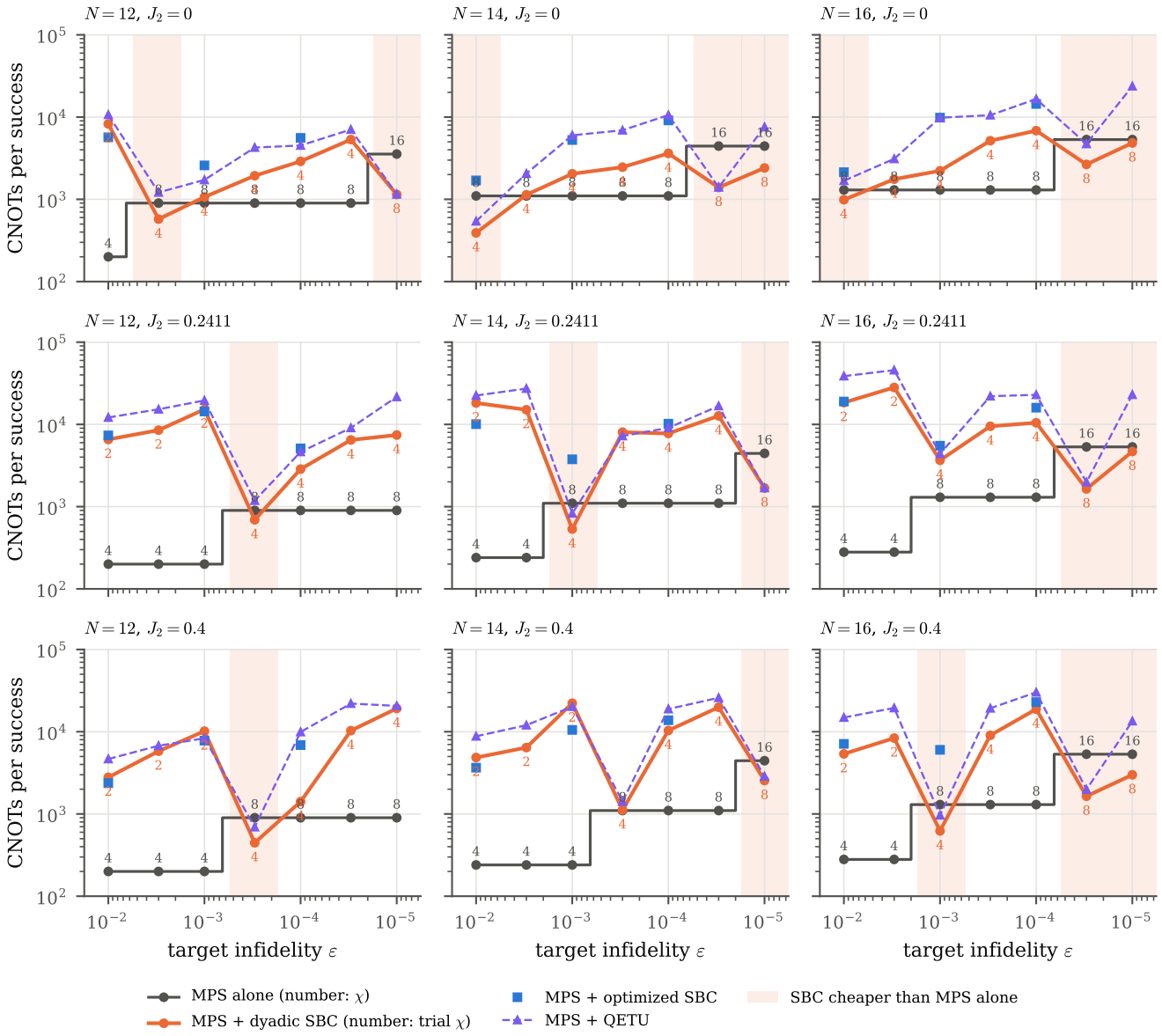


FIG. 12. Measured cheapest route to infidelity ϵ on chains of $N = 12, 14, 16$ (columns) at $J_2 = 0, 0.2411, 0.4$ (rows), all by exact simulation. Grey: cheapest MPS that meets $1 - \gamma \le \epsilon$ alone (number: χ). Orange: cheapest smaller MPS followed by the dyadic SBC filter (number: trial χ); blue squares: optimized SBC (designed at $\epsilon = 10^{-2}, 10^{-3}, 10^{-4}$ only); purple: QETU. Shaded: the SBC route is cheaper. Cost is (trial + filter CNOTs)/p. Sources: `results/emp_actual1d.jsonl` and, for $N = 16$, $J_2 = 0.2411$, `results/emp_actual.json`.

by a factor of four per bond-dimension step, and its cost in the overlap is the logarithm of Sec. IV. On the 16-site chain in the same comparison the filter wins only at $\epsilon = 3\times10^{-5}$ and 10^{-5}. The result depends on the MPS circuit cost, which we take from the generic quantum-Shannon bound. Table VIII re-costs every MPS circuit, in both routes, with a factor $f \le 1$ and with the lower bound for isometries from b to $b+1$ qubits [37], $\lceil (3 \cdot 4^b - b - 2)/4 \rceil$ CNOTs per site (about 0.4 of the bound used), which no compiler of these site unitaries can beat. At that lower bound the filter still wins 24 of the 28 cylinder targets, and it keeps a majority until MPS circuits are about eight times cheaper than the bound we use; on chains its 18 wins fall to 11. Over these targets the cylinder result is therefore robust to the MPS compiler; the chain result is not. These are small cylinders, at sizes where exact simulation is possible, and we leave a systematic two-dimensional study to future work; with certified plans in place of measured ones, the same comparison extended to 4×12 finds the filter to be the only route at $\epsilon = 10^{-4}$ on the larger cylinders. For the two-dimensional J_1–J_2 model, resource estimates for phase estimation have

TABLE VI. Gap and overlap overestimation for every $N \in \{8, 10, 12, 14\}$ and J_2 (12 instances), $\epsilon = 10^{-2}$, second order: number of plans that miss the target and the worst measured infidelity. f: assumed over true gap or γ. Source: `results/ext_sensitivity.json`.

		dyadic		optimized		QETU	
input	f	miss	worst	miss	worst	miss	worst
gap	1	0/12	1.7×10^{-3}	0/12	4.2×10^{-3}	0/12	2.3×10^{-3}
gap	1.1	0/12	6.5×10^{-3}	1/12	1.1×10^{-2}	1/12	1.0×10^{-2}
gap	1.25	0/12	6.8×10^{-3}	5/12	2.6×10^{-2}	5/12	3.0×10^{-2}
gap	1.5	2/12	6.3×10^{-2}	7/12	6.8×10^{-2}	7/12	9.2×10^{-2}
gap	2	8/12	1.4×10^{-1}	7/12	1.5×10^{-1}	11/12	2.2×10^{-1}
gap	3	11/12	2.2×10^{-1}	12/12	2.5×10^{-1}	12/12	3.1×10^{-1}
γ	1.02	1/12	2.4×10^{-2}	1/12	2.4×10^{-2}	1/12	2.4×10^{-2}
γ	1.05	3/12	5.5×10^{-2}	3/12	5.5×10^{-2}	3/12	5.5×10^{-2}

assumed the state preparation to be negligible [29]; these results suggest that a compiled low-χ MPS plus a short filter is a practical way to meet that assumption.

VII. COMPARISON WITH A CHEBYSHEV FILTER (QETU)

To compare with a polynomial filter under the same accounting, the pipeline also designs the quantum eigenvalue transformation of unitaries (QETU) of Dong, Lin and Tong [9]. It uses the same primitive, d queries of $e^{-i\tau H_s \otimes Z_a}$, the controlled-reversal form also used by the rodeo algorithm [10], but interleaves them with ancilla rotations and post-selects only once. On every eigenvector the ancilla then undergoes quantum signal processing in $x = \cos(\tau E)$, so the ancilla-0 block is $p(\cos \tau H_s)$ for any real even polynomial p of degree d with $|p| \le 1$ on $[-1, 1]$ [9]. With $\tau = \pi/2$ the scaled spectrum maps to $x \in [0, 1]$ and the stop band to $[0, x_s]$, $x_s = \cos(\pi\Delta/2)$. We take the Chebyshev filter

$$p(x) = \frac{T_d(x/x_s)}{T_d(1/x_s)}, \qquad \eta = \frac{1}{T_d(\cos(\tau w_0)/x_s)}, \tag{15}$$

which is extremal: no polynomial of degree d bounded by 1 is smaller on the stop band relative to its value at the pass band. It is certified in closed form (p is monotone on $[x_s, 1]$ and equioscillates on $[-x_s, x_s]$), has $p(0) = 1$ so $p_g = \gamma F_{\text{pass}}^2 \approx \gamma$, and needs $T = d\tau \approx \Delta^{-1} \ln(2/\eta)$, two to three times less than the dyadic ladder. The Trotter bound, step allocation and Theorem 1 carry over unchanged, with one pulse per query and two ancilla rotations per query (the phase and the identity term). The QSP phases are computed classically; our cost model does not include that step. For second-order formulas $S_2(-\tau) = S_2(\tau)^\dagger$, so the Trotterized circuit implements exactly $p(\tilde{X})$ with $\tilde{X} = \frac{1}{2}(S_2(\tau) + S_2(-\tau))$, which is how we simulate it; the test suite checks this against an explicit ancilla circuit with QSP phases.

Certified. The Chebyshev QETU filter, planned with the same Trotter, synthesis and cost model, is feasible in all 36 second-order cases (0 failures) and is a median 3.3× (range 2.1–3.9) cheaper than the dyadic schedule. Against the optimized SBC design, where that design exists, the ratio is a median 1.00 (range 0.85–4.3, 29 cases): the two cost the same to within the spread, because both reach $T \approx \Delta^{-1} \ln(1/\eta)$ up to a constant and the Trotter cost depends only on T. At $N = 48$, $J_2 = 0.2411$ QETU needs 8.2×10^7 CNOTs per success against 2.7×10^8 for the dyadic plan (Fig. 9). Simulated as in Sec. III B, the certificate held in 24 of 24 QETU plans ($N \le 14$, both targets), with a median looseness of 7.5×; its leakage term is tight (median 1.3×) because the Chebyshev filter equioscillates across the stop band. In the controlled study of Sec. IV, the certified QETU plan is a median 0.30 times the certified dyadic one and cheaper in all 222 cases, and it is the cheapest certified filter in 36, 49 and 74 of 74 cases at $\epsilon = 10^{-2}$, 10^{-3}, 10^{-4} (the optimized SBC design takes the rest). Certified, the SBC filter therefore offers no cost advantage over a polynomial filter; what it offers is the absence of a phase-finding step and a measurement after every pulse, which an early-abort protocol (not modeled here) could exploit.

Measured. The gap/overlap study simulates both filters on every case (Figs. 4–6, bottom rows and purple points). Measured, the dyadic SBC filter is the cheaper one at loose targets: QETU is cheaper in 0 of 74 cases at $\epsilon = 10^{-2}$, 9 of 74 at 10^{-3}, but 56 of 74 at 10^{-4}; over all cases QETU costs a median 1.26× the dyadic filter. The reason is the Trotter step, not the filter. QETU needs less evolution time (a median 0.65× the dyadic time), but it must take one fine Strang step per short query, $\delta t = \tau/W \propto 1/N$, so its step count is a median 1.40× the dyadic one and its CNOTs grow as $N^{2.22}$ against $N^{1.30}$. The long dyadic pulses tolerate steps of order $1/J_1$, but these shrink as roughly $\epsilon^{0.3}$, and by $\epsilon = 10^{-4}$ the shorter QETU filter wins. QETU's measured cost is also the more regular of the two (Table IV): its fit holds to 1.04×, because its degree grows by two at a time where the dyadic ladder grows in whole pulses.

A. What the Chebyshev filter needs

To see how the gap and the overlap act on the Chebyshev filter itself, we replace the certificate by simulation for $N = 8$–16 (675 cases). Each trial state is decomposed on the exact eigenbasis (dense diagonalization per S^z block), so the smallest QETU degree $d_{\min}$ that reaches ϵ for that state is found exactly; at that degree we then find the smallest number of Strang steps per query with the Trotterized circuit simulated on the full state vector, allowing a slightly higher degree when it makes one step per query enough. Three families separate the inputs. In the first, $J_2 = 0.2411$ for every N and the gap

TABLE VII. Overlap γ and expected CNOTs per success for trial bond dimension $\chi = 1, 2, 4, 8$ ($\epsilon = 10^{-2}$, second order, cheaper schedule; no filter when $1 - \gamma \le \epsilon$).

N	J_2	γ_1	CNOT	γ_2	CNOT	γ_4	CNOT	γ_8	CNOT
16	0	0.028	2.8×10^{7}	0.475	2.3×10^{5}	0.984	2.5×10^{4}	1.000	1.3×10^{3}
16	0.2411	0.019	1.3×10^{8}	0.703	3.5×10^{5}	0.998	2.8×10^{2}	1.000	1.3×10^{3}
16	0.4	0.000	7.6×10^{10}	0.911	1.5×10^{5}	0.999	2.8×10^{2}	1.000	1.3×10^{3}
32	0	0.001	2.8×10^{10}	0.234	1.7×10^{7}	0.913	4.3×10^{5}	0.998	2.9×10^{3}
32	0.2411	0.001	2.5×10^{11}	0.373	2.9×10^{7}	0.972	7.7×10^{5}	0.999	2.9×10^{3}
32	0.4	0.000	1.7×10^{17}	0.768	2.3×10^{6}	0.990	6.0×10^{2}	1.000	2.9×10^{3}
48	0	0.000	5.5×10^{12}	0.125	1.7×10^{8}	0.817	2.8×10^{6}	0.990	4.5×10^{3}
48	0.2411	0.000	9.0×10^{13}	0.191	2.7×10^{8}	0.920	4.8×10^{6}	0.995	4.5×10^{3}
48	0.4	0.000	3.2×10^{23}	0.640	2.4×10^{7}	0.976	1.6×10^{6}	0.998	4.5×10^{3}

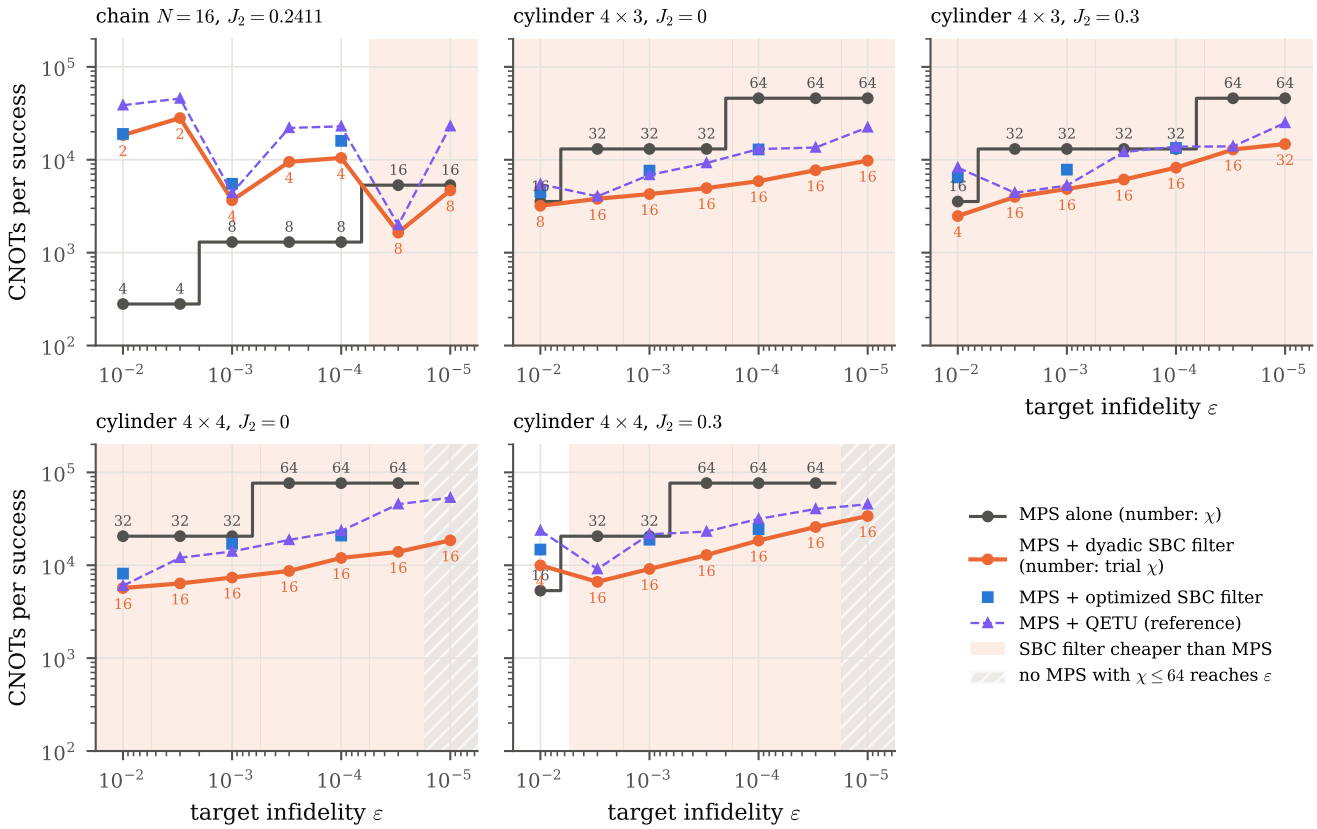


FIG. 13. Measured cheapest route to infidelity ϵ on the 16-site chain and on small cylinders. Grey: the cheapest MPS that meets the target alone (number: χ). Orange: the cheapest smaller MPS followed by the dyadic SBC filter (number: trial χ); blue squares: optimized SBC; purple: QETU. Shaded: the filtered route is cheaper; hatched: no MPS with $\chi \le 64$ reaches ϵ. Every cost is (trial + filter CNOTs)/p from exact simulation. Source: `results/emp_actual.json`.

closes with N. In the second, J_2 is chosen at each N (by bisection on ED) so that the gap is $E_1 - E_0 = 0.40$ exactly ($J_2 = 0.384, 0.510, 0.539, 0.550, 0.555$ for $N = 8$–16). In the third, J_2 is tuned at $N = 10$, 12 and 14 to gaps from 0.30 to 0.55. For the trial states we use the optimal MPS at $\chi = 1$, 2, 4 and synthetic states $\sqrt{\gamma}|E_0\rangle + \sqrt{1-\gamma}|\xi\rangle$ with γ from 0.03 to 0.9, ξ being the normalized excited part of the $\chi = 2$ MPS, so that the overlap changes while the shape of the excited weight does not. $\epsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$.

The degree is set by the gap and the target alone. The

TABLE VIII. Robustness of the measured crossovers to the cost of the MPS circuits: targets (of 28 cylinder and 63 chain targets) at which MPS + SBC filter beats the MPS alone when every MPS circuit is re-costed by a factor f, with the quantum-Shannon bound used throughout and with the isometry lower bound [37].

MPS cost model	$f=1$	1/2	1/4	1/8	1/16
QSD bound (used), cylinders (of 28)	27	25	23	19	13
QSD bound (used), chains (of 63)	18	13	6	4	0
isometry lower bound, cylinders (of 28)	24	22	18	12	9
isometry lower bound, chains (of 63)	11	6	4	0	0

measured degrees collapse onto

$$d_{\min}\,\Delta \;\simeq\; \frac{2}{\pi}\cosh^{-1}\frac{1}{\eta}, \qquad \eta^2=\frac{\gamma\epsilon}{1-\gamma}, \tag{16}$$

for every family, trial and target (Fig. 14b). This is the certified degree with the true gap: the certificate asks for a median 1.09 times $d_{\min}$. Where the excited weight of the trial sits does not help. The Chebyshev filter is equiripple, so every state above the gap is suppressed by the same factor $1/T_d(1/x_s)$, and a trial whose excited weight lies far above the gap needs the same degree as one whose weight sits at the gap. Over the gap scan $d_{\min}\propto\Delta^{-1.00}$.

The certificate's Trotter count is pessimistic, not its filter. One Strang step per query was enough in all 675 cases, against a certified median of 16 steps per query. Commutator bounds are worst-case over all states, but the filter only has to act well on a state supported near the bottom of the spectrum. As a result the certified cost is a median 22 times what the filter needs (range 7–2×10^2), and the real cost follows the degree:

$$\mathrm{cost}_{\mathrm{real}} \;\approx\; \frac{c_{\mathrm{step}}(N)}{\gamma}\,d_{\min} \;\propto\; \frac{c_{\mathrm{step}}(N)\,W}{g\,\gamma}\,\ln\frac{1-\gamma}{\gamma\epsilon}, \tag{17}$$

with c_{step} the CNOTs of one step ($\propto N$) and $g=E_1-E_0$. Fitting the simulations gives overlap $\gamma^{-0.97}$ once the logarithm is divided out ($\gamma^{-1.21}$ as a plain power, against $\gamma^{-1.45}$ for the certified plan) and gap $\Delta^{-0.98}$ (certified $\Delta^{-1.38}$). With the gap held at 0.40 the cost grows as $N^{2.3}$; at fixed J_2, where g also closes, as $N^{3.1}$ (Fig. 14c). The difference is the $1/g$ of Eq. (17).

Gap and overlap are not independent inputs. Tuning J_2 to hold the gap moves the chain toward the dimerized point $J_2=1/2$, where a $\chi=2$ MPS is close to exact: at the gap-matched couplings the $\chi=2$ overlap is 0.97–1.00 at every N, against 0.70–0.90 at $J_2=0.2411$. Up to $J_2\approx1/2$ a larger gap and a better trial come together, so for real trials the two effects compound; beyond it the gap keeps growing while the $\chi=2$ overlap falls again (to 0.78 at $J_2=0.68$, $N=14$). Holding one of them fixed through J_2 therefore always moves the other, which is why the scans here use synthetic trials and Sec. IV a sublattice rotation of a real trial circuit.

VIII. MEASURED VERSUS CERTIFIED

Every comparison above has two versions, and they do not always agree (Table IX). The measured cost is not implementable as it stands: it picks the shortest filter and the coarsest Trotter step that reach ϵ for the given trial, which needs the exact ground state. It is the cost an ideal a-posteriori procedure could approach, and it shows what the circuit actually has to do. The certified cost is what the pipeline plans from Theorem 1 without that knowledge, and can be run as is. The two differ mainly through the Trotter step (Sec. IV): the certified step is 60–75 times finer, and its overhead grows with the filter length.

This difference decides which filter is cheapest. Measured, the long SBC pulses tolerate steps of order $1/J_1$ and win at loose targets; certified, every filter pays the same worst-case step per unit time, the shorter Chebyshev filter wins more often, and the optimized SBC design is the only one that competes. It also decides the end-to-end comparison. Certified, the filtered route is 1–2 orders of magnitude more expensive than measured. Against compiling a better MPS it loses at all 63 chain targets (median 152.3× the MPS cost, never below 13.5×) and at every cylinder target where some $\chi\le64$ reaches ϵ (median 5.2×), with one exception: QETU on 4×3, $J_2=0$, at $\epsilon=10^{-4}$, which is 1.2× cheaper and where the SBC filter ties. It remains the only route where no MPS reaches the target. The measured cylinder advantage is therefore a statement about what the circuit needs, which an implementation can claim only with a tighter error bound, for example one restricted to the low-energy subspace [19, 31], or with an a-posteriori check of the Trotter step (shrinking it until a measured energy or fidelity proxy stops changing). Two smaller oracle elements are the exact gap and overlap used to design the measured filters; the sensitivity study of Sec. V B shows that an overestimated gap is the dangerous error, so a margin on E_1 is needed in practice.

IX. DISCUSSION

What makes the filter advantageous. The controlled study separates three regimes. (i) The gap determines the coherent circuit: evolution time $\propto L/\Delta$, independent of N at fixed gap, and CNOTs and R_z per shot that scale as a power of $1/\Delta$. A system with a large gap in absolute units is the natural target, and the filter's depth is what hardware with a coherence or logical-error budget has to accommodate. (ii) The trial overlap enters the circuit only through a logarithm and the expected cost through $1/\gamma$: the filter is tolerant of a poor trial, paying in repetitions rather than in depth. This is where it has

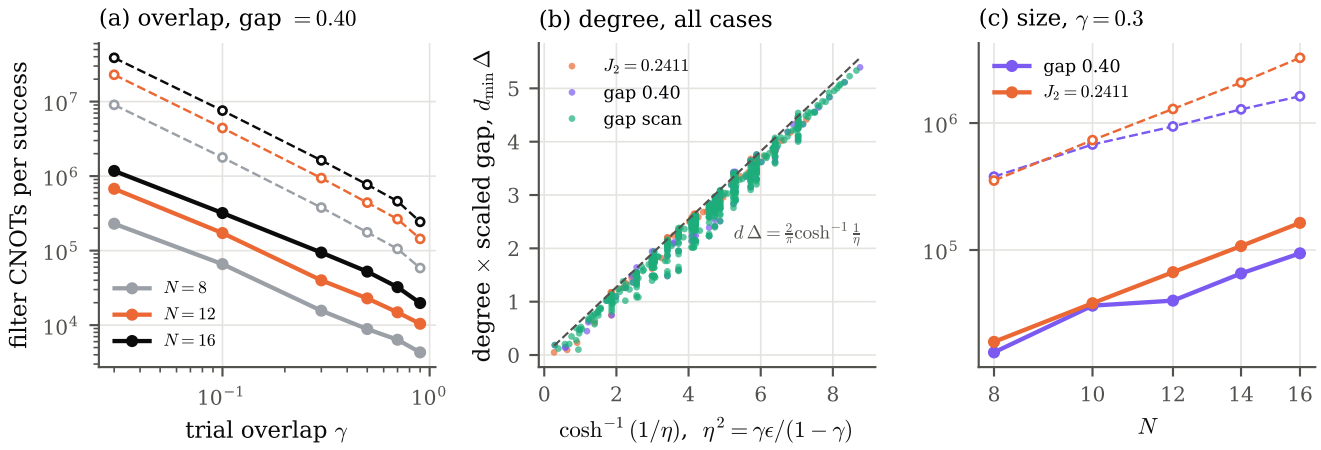


FIG. 14. What the filter actually needs, from exact spectra and state-vector simulation ($N = 8$–16, QETU, second order). (a) CNOTs per success versus overlap at a gap held at 0.40, $\epsilon = 10^{-3}$: simulated (filled) and certified (open, dashed). (b) Smallest degree times scaled gap for all 675 cases against $\cosh^{-1}(1/\eta)$; dashed: Eq. (16). (c) Cost versus N at $\gamma = 0.3$, $\epsilon = 10^{-3}$, with the gap held at 0.40 or J_2 held at 0.2411. Source: `results/emp_trotter.json`, `sbpipe_study/run_empirical.py`.

TABLE IX. Each comparison measured (shortest filter and coarsest Trotter step that reach ϵ in exact simulation; uses the ground state) and certified (planned from Theorem 1; implementable). Controlled study: 74 cases per target. Crossovers: targets 10^{-2}–10^{-5}; certified SBC uses the pipeline's automatic choice of dyadic or optimized, QETU in parentheses. Sources: `results/gapov*.jsonl`, `results/emp_actual*.json*`, `results/cert_crossover.jsonl`.

comparison	measured	certified
Cheapest filter, $\epsilon = 1 \times 10^{-2}$ (dy/opt/QETU)	23/51/0	0/38/36
Cheapest filter, $\epsilon = 1 \times 10^{-3}$ (dy/opt/QETU)	16/57/1	0/25/49
Cheapest filter, $\epsilon = 1 \times 10^{-4}$ (dy/opt/QETU)	5/32/37	0/0/74
MPS + filter beats MPS alone, 9 chains	18 of 63	0 of 63 (QETU 0)
MPS + filter beats MPS alone, 4 cylinders	27 of 28	2 of 28 (QETU 3)

an edge over compiling a better trial, whose cost grows with the bond dimension needed to reach ϵ directly. (iii) The target enters through L and, for SBC, through the Trotter step, $\delta t \propto \epsilon^{0.3}$; tight targets are where a compiled MPS runs out first. For the chain, the MPS infidelity falls quickly with χ, so the filter pays off only as a last-mile polisher, on a trial within a factor of about 8.8 of the target (Sec. V D); for cylinders, where a snake MPS struggles, it pays off at 27 of the 28 targets we measured (Sec. VI).

Dyadic, optimized or Chebyshev. Both SBC schedules beat the Chebyshev filter at $\epsilon \geq 10^{-3}$, measured (certified, the Chebyshev filter is usually the cheaper one, Sec. VIII), because their long smooth pulses tolerate Trotter steps of order $1/J_1$; the Chebyshev filter is shorter and catches up at $\epsilon = 10^{-4}$. The optimized SBC design is the cheapest of the three at $\epsilon \leq 10^{-3}$ in most cases, measured (median 0.89× dyadic) and even more so certified (0.35×), but it needs a numerical design that occasionally fails and must be repeated; the dyadic ladder is the robust default and the natural fallback, which is how the pipeline's "auto" mode uses them.

Amplitude amplification. All costs here are repeat-until-success, C/p per success. Amplitude amplification [35] would instead run the filter coherently about $\pi/(4\sqrt{p})$ times, each round costing the trial and filter circuits and their inverses, for an expected cost near $(\pi/2)C/\sqrt{p}$: a saving of about $(\pi/2)\sqrt{p}$, i.e. worthwhile only for $p \lesssim 0.4$. At the lowest overlap of the controlled study ($p \approx \gamma = 0.03$) it would cut the cost per success by about 3.7×, at the price of a coherent circuit about six times deeper, which works against the depth budget the gap already sets. Two further costs apply to the cosine filter. The SBC circuit measures the ancilla after every pulse; a coherent version needs one ancilla per pulse (or the pulses combined into a single post-selection, as in QETU), and with p known only as an estimate a fixed-point variant [36] is needed, which costs a further logarithmic factor. Where the filter wins in the crossovers, the trial is good and $p \geq 0.77$, so amplification would not help; it matters in the low-overlap regime, where it changes the $1/\gamma$ of the cost laws into $1/\sqrt{\gamma}$. An early-abort protocol that stops at the first failed pulse is the cheaper complement for SBC and is likewise not modeled.

What is guaranteed. Theorem 1 holds whenever its inputs are right. Of these, the spectral upper bound (Weyl) and the filter certificate are rigorous, and the

Trotter constants are exact up to the Lanczos tolerance of their largest local norms. The interval for E_0, the lower bound on E_1 and the overlap γ come from DMRG and are well-converged estimates, not bounds; for $N \le 14$ ED confirms them, beyond that the certificate is conditional on all three. The gap is the critical input: an overestimate of 10–25% already broke the target for the optimized and Chebyshev filters (Table VI). Rigorous lower bounds on E_1 (e.g. Temple-type bounds) would remove the gap condition, and with them the Eckart inequality $1-\gamma \le (\langle H\rangle_\psi - E_0)/(E_1 - E_0)$ would remove the overlap condition.

Where the certified cost goes. The certified filter length is within a factor of about two of what simulation needs; the certified Trotter step is 60–75 times finer than needed, and the certified step count grows as L^2 rather than L. Tighter Trotter bounds, for example restricted to the low-energy subspace [19, 31], or a posteriori error estimates, would recover most of the 62-fold median gap between certified and measured cost; a better filter design would not.

Limitations. Noiseless simulation; repeat-until-success rather than early abort; all-to-all connectivity (a nearest-neighbor layout would add SWAPs, mostly for the ancilla, which couples to every bond); trial circuits costed by a generic decomposition bound rather than compiled; exact simulation and the controlled study only up to $N = 14$–16, so the fitted exponents are descriptive; the gap range that can be held fixed across N is narrow (0.30–0.55); E_1 is taken as the lowest $S^z = 1$ state, which is the first excitation in every ED case here but an assumption for larger N.

CODE AND DATA AVAILABILITY

The `sbpipe` package, the study scripts, all raw results (`sbpipe_study/results/*.json`) and the sources of this paper are available at https://github.com/aaronseymour7/FilterPipeline.

Appendix A: Proof of Theorem 1

Write the trial state as $|\psi\rangle = \sqrt{\gamma'}|g\rangle + \sqrt{1-\gamma'}|\chi\rangle$ with $|g\rangle$ in the ground space (energy $e_0 \in [0, w_0]$ in scaled units), $|\chi\rangle$ orthogonal to it and $\gamma' \ge \gamma$.

Lemma 1 (leakage)**.** *For the exact filter, $v = F(H_s)|\psi\rangle$ has weight outside the ground space at most ℓ, and $\|v\|^2 \ge p_g$.*

Proof. $F(H_s)$ is a function of H_s, so $v = \sqrt{\gamma'}F(e_0)|g\rangle + \sqrt{1-\gamma'}F(H_s)|\chi\rangle$ with the two parts orthogonal. Every component of $|\chi\rangle$ has energy in $[\Delta, 1]$, so $\|F(H_s)\chi\| \le \sup_{[\Delta,1]}|F| \le \eta F_{\text{pass}} \le \eta|F(e_0)|$. The excited weight is $x/(a+x)$ with $a = \gamma' F(e_0)^2$, $x = (1-\gamma')\|F\chi\|^2 \le (1-\gamma')\eta^2 F(e_0)^2$; it is increasing in x and, after dividing by $F(e_0)^2$, decreasing in γ', which gives ℓ. Finally $\|v\|^2 \ge a \ge \gamma F_{\text{pass}}^2$. □

For pure states the trace distance is $\sin\angle$, so $D(\hat{v}, |g\rangle) = \sqrt{\ell'} \le \sqrt{\ell}$ ($\hat{v} = v/\|v\|$).

Lemma 2 (implementation error)**.** *Let $B_i = \frac{1}{2}(e^{-i\phi_i}\tilde{S}_i(t_i) + e^{i\phi_i}\tilde{S}_i(-t_i))$ be the ancilla-0 block of the implemented pulse and $C_i = \cos(H_s t_i + \phi_i)$. Then $\|\prod_i B_i - \prod_i C_i\| \le \epsilon_U$.*

Proof. $C_i = \frac{1}{2}(e^{-i\phi_i}e^{-it_iH_s} + e^{i\phi_i}e^{it_iH_s})$, so $\|B_i - C_i\| \le \frac{1}{2}(\|\tilde{S}_i(t_i) - e^{-it_iH_s}\| + \|\tilde{S}_i(-t_i) - e^{it_iH_s}\|) \le \epsilon_i$, where ϵ_i is the product-formula bound (symmetric in the sign of t) plus the synthesis error of the pulse's rotations. Since $\|B_i\|, \|C_i\| \le 1$, telescoping gives $\|\prod B_i - \prod C_i\| \le \sum_i \epsilon_i$. A synthesized trial circuit prepares $|\tilde{\psi}\rangle$ with $\|\tilde{\psi} - \psi\| \le R_{\text{trial}}\delta$, and $\|\prod B_i\| \le 1$, so the total error of the unnormalized output is at most ϵ_U. □

With $\tilde{v} = \prod B_i|\tilde{\psi}\rangle$, $\|\tilde{v} - v\| \le \epsilon_U$. The distance from v to the line through $\tilde{v}$ is at most $\|v - \tilde{v}\|$, so $\sin\angle(v, \tilde{v}) \le \epsilon_U/\|v\| \le \epsilon_U/\sqrt{p_g}$. The triangle inequality for the trace distance gives the bound on D. The states $\hat{v}$, $|g\rangle$ and the normalized output are pure, so $D^2 = 1 - |\langle g|\tilde{\psi}_{\text{out}}\rangle|^2$; since $|g\rangle$ lies in the ground space, $\text{Tr}(\Pi_0\tilde{\rho}) \ge |\langle g|\tilde{\psi}_{\text{out}}\rangle|^2$, which gives (8). (For a mixed output, e.g. under noise, a trace-distance bound alone would not bound the ground-space weight by D^2; this case is outside the theorem.) The success probability is $\|\tilde{v}\|^2 \ge (\|v\| - \epsilon_U)^2 \ge (\sqrt{p_g} - \epsilon_U)^2$. For the energy, the normalized state has weight at least $1 - D^2$ on the ground space and at most D^2 on levels $\le E_{\text{top}}$, so $\langle H\rangle - E_0 \le (E_{\text{top}} - E_0)D^2$. □

Appendix B: Reproducing the results

The code, raw results and paper sources are at https://github.com/aaronseymour7/FilterPipeline. After `pip install -r requirements.txt`, all commands below run from the repository root. `python sbpipe_study/run_study.py all` runs the DMRG front end (cached), the validation (A), scaling (B), sensitivity (C), ansatz (D) and oracle (E) studies and writes `results/*.json`; `python sbpipe_study/make_figures.py` writes every figure, table and the numerical macros used in this text. The stored results were produced with per-term CNOT counts and converted to the bond compile of Eq. (9) by `sbpipe_study/recompile_results.py`, which checks every stored count against the per-term formula before replacing it; a fresh run counts bonds directly. `python -m pytest sbpipe/tests` runs the component checks (Pauli algebra, Weyl bound, Trotter bounds against dense matrix exponentials, the Dirichlet identity (3), filter certificates against a dense grid, gate counts against an explicit merged sequence and against qiskit, and an end-to-end certificate check).

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